

Doctoral Dissertation

Machine Learning for Precision Therapy of Triple-Negative Breast Cancer: Exploring the predictive ability of blood cytokines for chemotherapy

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Abstract

Triple-negative breast cancer (TNBC) is a highly aggressive type of breast cancer which does not express estrogen or progesterone receptors and does not overexpress the HER2 protein. Currently, treatment options for TNBC are limited. Recent research indicates that cytokines play roles in TNBC tumor growth and metastasis, and that chemotherapy can modulate the expression of cytokines. Thus, we hypothesized that cytokines may be useful for predicting the outcomes of chemotherapy of TNBC patients.

To test this hypothesis, we obtained cytokine measurements from blood samples of TNBC patients at multiple time points. We analyzed this data using Weighted Gene Co-expression Network Analysis (WGCNA) to find cytokines correlated with chemotherapy treatment outcomes, and calculated q-values to control the false discovery rate. We discovered five cytokines, IL-1 α , TRAIL, SCF, RANTES, and IL-16, which were correlated with good chemotherapy outcomes. We performed protein-protein interaction analysis, which indicated that these cytokines are involved in immune response. We then performed logistic regression with L2 regularization, which indicated that IL-1 α and TRAIL can reliably predict the outcomes of eribulin chemotherapy. These results indicate that cytokine expression levels in the blood may be effective tools in predicting the outcome of eribulin chemotherapy in TNBC patients.

Keywords:

machine learning, oncology, breast cancer, cytokines, chemotherapy, precision medicine

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1. Introduction

1.1 An overview of triple-negative breast cancer

Triple-negative breast cancer (TNBC) is a type of breast cancer characterized by a lack of expression of the estrogen and progesterone receptors, and lack of overexpression of the HER2 protein [1]. TNBC tumors are highly aggressive, often having moderate to high grade [2]. It is estimated that TNBC comprises about 10-15% of breast cancer cases [2]. African American women and persons with germline *BRCA* and *PALB2* mutations are at greater risk of TNBC [3].

Due to the nature of TNBC (i.e., lacking expression of the estrogen and progesterone receptors, and lacking overexpression of HER2 protein), this subtype of breast cancer does not have obvious molecular targets. Because of this, TNBC treatment options are currently limited to surgery, chemotherapy, and radiotherapy [4]. Since treatment options are limited, treatment of TNBC is more difficult than other types of breast cancer, and prognosis is often poorer than other types of breast cancer [2]. In contrast, other types of breast cancer, such as those that are positive for the estrogen receptor or positive for HER2 overexpression, can be treated by various agents tailored to these molecular targets [2]. Due to the high aggressiveness of this type of cancer, and the limited number of treatment options, it is of great importance to develop new methods that can aid in the treatment of TNBC.

1.2 The role of cytokines in immune response to cancer

The immune system is the human body's natural protection against cancer. Some immune cells, such as natural killer (NK) cells and cytotoxic T-lymphocytes, are able to kill malignant cells and inhibit tumor growth and metastasis [5][6]. The function of the immune system can be influenced by a variety of factors which can stimulate or inhibit the immune response. One example of a factor that can influence the function of the immune system is a class of proteins known as cytokines. Cytokines are proteins which play important roles in mediating and regulating the immune system [7]. Due to their immunoregulatory function, cytokines have been considered as a potential tool for cancer therapy. Previous research has indicated

that some cytokines have anti-tumor effects [8]. Such anti-tumor effects can be achieved in a variety of ways, such as by enhancing the immune system's ability to kill cancer cells, or by inhibiting cancer-induced immunosuppression [8]. Because of these anti-tumor effects, the use of cytokines as a method of anti-cancer therapy is an active area of research [8].

1.3 Potential application of cytokines for triple-negative breast cancer therapy

Different types of cancer have different characteristics. Hence, a therapy or drug that is effective for one type of cancer may not necessarily be effective for another type of cancer. It is therefore necessary to investigate the utility of each treatment method for the specific type of cancer to be treated.

Previous research has indicated that cytokines play roles in TNBC, and may potentially be useful tools for the treatment of this disease [9]. For example, the cytokines interleukin (IL)-6, IL-8, and Chemokine (C-C motif) Ligand 5 (CCL5) have been shown to play significant roles in TNBC growth and metastasis [9]. Previous research also indicates that cytokines in the blood are important sources of clinical information [10][11]. Cytokine expression can be affected by various factors, such as the presence or absence of diseases, and the type of treatment used (e.g., chemotherapy) [12]. For example, in a study by Groysman et al. (2021), the cytokines Chemokine (C-X-C motif) Ligand 9 (CXCL9) and IL-6 were upregulated by doxorubicin, a widely used chemotherapy drug [12]. Cisplatin, another widely used chemotherapy drug, was shown to downregulate IL-17 in the same study [12].

These studies suggest that cytokines may be useful tools for TNBC therapy. Since cytokines are known to play roles in TNBC growth and metastasis, and may be upregulated or downregulated by chemotherapy, it is natural to ask questions such as “what is the significance of cytokines in TNBC chemotherapy? Which cytokines are affected by commonly-used TNBC chemotherapy drugs? Is it possible to use cytokines as a tool for TNBC treatment?”. In this study, we aimed to answer such questions using a combination of clinical trial data, statistical analysis, and machine learning.

1.4 Our approach – Analysis of time series cytokine data of TNBC patients receiving chemotherapy

When investigating how the human body responds to disease and treatment, it is important to consider not only the status of the body at a single point in time (e.g., at the time of diagnosis), but also how the status of the body changes with time. For example, suppose that we observed the status of a patient not only at the time of diagnosis, but also at various other time points during treatment. Such observations would give us valuable information that would aid in our efforts to understand the mechanisms of disease and how the body responds to treatment. Furthermore, since different people have different responses to disease and treatment, such information would be useful in designing an optimal treatment strategy that is well-suited for that particular patient. Not all patients respond the same way to chemotherapy. TNBC is known to have multiple subtypes that have different gene expression profiles, and patient responses to chemotherapy can differ significantly between these subtypes [13][14][15]. Also, chemotherapy is known to cause adverse side effects in patients [16]. Due to the differences in patient response to chemotherapy, and the negative side effects of chemotherapy, it would be very helpful and important to have a method of predicting whether or not a certain chemotherapy drug would be effective in treating a particular patient.

In this study, we analyze cytokine data from TNBC patients at multiple time points. The data from this study comes from a randomized phase II neoadjuvant study, the Japan Breast Cancer Research Group (JBCRG)-22 trial. In this clinical trial, time-series sampling of patients' blood was performed. The blood samples were collected at five different time points: at diagnosis (pre-treatment), cycle 1 of chemotherapy, cycle 4 of chemotherapy, pre-surgery, and 1 year after surgery. Cytokine levels were measured from these blood samples. Furthermore, data about cancer-intrinsic features, such as homologous recombination deficiency (HRD) scores and the presence of BRCA mutations, were also collected. Clinical data of patients (e.g. age at diagnosis, tumor size) were also collected. These datasets were used to search for correlations between cytokine profiles and clinical characteristics. We analyzed this data set (time-series cytokine datasets, patient information, clinical outcomes, and intrinsic features of cancers) using correlational analyses in order to identify correlations

between cytokines, patient characteristics, clinical outcomes, and cancer-intrinsic features. We also performed logistic regression analysis in order to predict treatment outcomes based on pre-treatment cytokine expression levels. This study highlights the utility of correlation analyses and logistic regression using datasets of various cytokines. The results of this study may be helpful in improving the precision of TNBC treatment.

2. Materials and methods

2.1 Study design

We obtained the data used in this study from a clinical trial performed by Masuda et al. (2021) [17]. This clinical trial was known as JBCRG-22. In this trial, a cohort of TNBC patients was randomized and divided into two groups (A and B) based on age, HRD scores, and the type of *BRCA1/2* pathogenic germline variant (PGV), if it was available. Patients less than 65 years old with an HRD score ≥ 42 , and all patients with the *BRCA1/2* PGV, were placed into Group A. Patients less than 65 years old with an HRD score <42 , and all patients 65 years old and above except those with the *BRCA1/2* PGV, were placed into Group B. The HRD scores were measured by Myriad Genetics (Salt Lake City, UT, USA). The HRD assay is a custom hybridization capture panel that targets 54,091 SNPs evenly distributed across the genome. Whole genome next-generation sequencing libraries are hybridized to this panel to enrich for sequence spanning these SNP locations. The enriched library is then sequenced using a HiSeq 2500 next-generation sequencer (Illumina). The sequencing reads are used to generate allele-specific copy number profiles, and these profiles are used for the calculation of the HRD score. Group A was further subdivided into arms A1 and A2, and Group B was further subdivided into arms B1 and B2. Patients in arm A1 were treated with paclitaxel followed by an anthracycline-based regimen, and patients in arm A2 were treated with eribulin plus carboplatin followed by an anthracycline-based regimen. Patients in arm B1 were treated with eribulin plus cyclophosphamide, and patients in arm B2 were treated with eribulin plus capecitabine. This was followed by the next phase of treatment, which depended on the antitumor effects of the previously mentioned treatments. Based on the effects of the treatment assessed at cycle 4, the patients in Group A were either treated with an additional four cycles of 5-fluorouracil–epirubicin–cyclophosphamide (FEC) or doxorubicin–cyclophosphamide (AC) or discontinued chemotherapy and underwent surgery. In Group B, if the tumors showed good clinical response at cycle 4 of the treatment, then the patient would be treated with an additional two cycles of eribulin plus cyclophosphamide or capecitabine, and then would undergo surgery. However, if patients had stable disease or residual invasive tumors as determined by core biopsy after the 4th treatment cycle, the treatment was switched

to a “rescue regimen” consisting of four cycles of an anthracycline-based regimen (FEC or AC), followed by surgery. This procedure is summarized in the flowchart in Fig. 1A.

Initially, 99 patients were recruited. However, 38 patients were excluded because either they declined to participate or their blood samples were unavailable, resulting in a dataset consisting of 61 patients (Fig. 1b). Blood samples were collected at five time points: pre-treatment, cycle 1-day 15, cycle 4, pre-surgery, and 1 year after surgery (Fig. 1c, Table 1), and cytokine concentrations were measured from these blood samples, resulting in a cytokine concentration data set with five time points. The following clinical information was also recorded: presence or absence of eribulin, time courses, age, tumor size, node metastasis, tumor grade, HRD scores, and treatment outcomes. A summary of the clinical information of the cohort of TNBC patients is given in Table 2.

<i>Treatment group</i>	<i>Sample collection timing</i>				
<i>(N = sample size)</i>	Pre-treatment	Cycle 1-Day 15	Cycle 4	Pre-surgery	1 year post surgery
<i>A1 (N = 18)</i>	18	18	17	18	18
<i>A2 (N = 12)</i>	12	12	11	12	11
<i>B1 (N = 14)</i>	14	14	13	13	14
<i>B2 (N = 17)</i>	17	17	15	16	14

Table 1. The number of blood samples collected for each time point. Total number of samples: 294.

The study complied with the Declaration of Helsinki and the Ethical Guidelines for Clinical Research of the Ministry of Health, Labor, and Welfare of Japan. The study protocol was reviewed and approved by the institutional review board of each participating institution. All patients provided written informed consent at primary registration.

2.2 Cytokine and chemokine measurements

A series of 69 cytokines and chemokines were measured with a Bio-Plex multiplex assay system (Bio-Rad, Hercules, CA, USA) according to the manufacturer's instructions (Table B1). We used 48-plex and 40-plex kits, which measure 48 cytokines and 40 cytokines simultaneously, respectively, to analyze 69 cytokines: 19 cytokines were measured twice using these two different kits, which utilize different antibodies to measure the same cytokines. The list of cytokines in these two plex kits is provided in Table B2. The collected serum samples were stored at -80°C until use. The samples were then thawed, and 12.5 μl from each sample was subjected to cytokine measurements. To cover accuracy, in each experiment, two technical replicates (1st and 2nd measurements) were performed on the same cytokines on different days, and a standard control sample was used for precision control before the start of sample measurement.

2.3 Detection of cytokine modules and correlations with clinical data using WGCNA

After obtaining the cytokine data, we analyzed this data using Weighted Gene Correlation Network Analysis (WGCNA). WGCNA is a software package that uses graph theory and network analysis to infer gene regulatory networks and to detect correlations between modules (i.e., clusters) of genes or cytokines and clinical phenotypes [18]. Before performing WGCNA, the average of the 1st and 2nd set of measurements of each cytokine were calculated for every sample (Fig. A1). Two samples (identified in the raw data set by the ID numbers 3064_2 and 2005_3) were excluded from the analysis because the 2nd set of measurement data of the 40-plex kit consisted entirely of missing data for these two samples. We excluded all cytokines that contained missing values in either the 1st or 2nd measurements for at least one sample (i.e., only cytokines which contained no missing values were included in the analysis). After this filtering procedure, a total of 52 cytokines remained. The cytokines that passed this filtering procedure can be seen in Table B3. These cytokines were then analyzed using WGCNA. One of the main parameters to be specified by the user in WGCNA is the soft threshold parameter β [18]. Among various values of β tested, the value $\beta = 3$ was chosen because it resulted in the best balance between scale-free topology model fit and mean connectivity. The WGCNA function `blockwiseModules` was used for constructing an

unsigned topological overlap matrix to represent the connections between the nodes of the graph with the parameters set as follows: TOMType = “unsigned,” minModuleSize = 2, reassignThreshold = 0, mergeCutHeight = 0.25, numeric labels = TRUE, pamRespectsDendro = FALSE. Pearson correlation coefficients between the module eigengenes and clinical data (e.g., patient age, HRD score) were then calculated, the corresponding p -values were calculated using Student’s t -test, and q -values were calculated with GraphPad Prism 9.4.1 (GraphPad Software, San Diego, CA, USA; www.graphpad.com) using the method of Benjamini and Hochberg [19]. The threshold q -value was set at 0.1 (i.e., q -values less than 0.1 to be discoveries).

2.4 Statistical analysis of individual cytokines

Our data set contains both discrete clinical data (i.e., data that can be represented as a finite number of discrete categories, such as treatment outcome and presence or absence of eribulin in the chemotherapy regimen) and continuous clinical data (i.e., data represented as continuous numerical values, such as HRD score and patient age). For modules that were found to be correlated with discrete clinical data, we separated the patients into their corresponding groups based on the discrete clinical data category of interest (e.g., patients treated with eribulin versus patients not treated with eribulin, patients with a favorable or unfavorable treatment outcome). We used the Mann-Whitney U -test to calculate the p -values of the differences in the medians between the different categories and calculated the q -values using the method of Benjamini and Hochberg (1995) [19]. q -values less than 0.1 were considered to be discoveries.

For modules that were found to be correlated with continuous clinical data, we used linear correlation analysis to search for statistically significant correlations. Pearson correlation coefficients (r), p -values, q -values, and coefficients of determination (R^2) were calculated using GraphPad Prism version 9.4.1. q -values below 0.1 were considered to be significant.

Table 2. Clinical characteristics of the patients in the various treatment arms

Characteristic	Treatment Arm			
	A1 (N = 18)	A2 (N = 12)	B1 (N = 14)	B2 (N = 17)
Age at secondary registration (years), median (range)	45.5 (28–64)	52.0 (31–63)	64.0 (35–70)	59.0 (43–70)
Age group at secondary registration (years), n				
<50	11	5	2	1
≥50	7	7	12	16
<65	18	12	7	10
≥65	0	0	7	7
Anthracycline treatment, n				
Yes	17	11	6	6
No	1	1	8	11
TNM classification: T (primary lesion), n				
T1c	9	4	1	4
T2	8	6	12	12
T3	1	2	1	1
Tumor size (primary lesion), mm				
Median (range)	21.0 (13–60)	30.0 (13–64)	30.0 (20–70)	25.0 (12–53)
<30.5, n	11	6	8	10
≥30.5, n	7	6	6	7
TNM classification: N (regional lymph node), n				
N0	12	8	8	11
N1	6	4	6	6

Histological grade (B&R), n				
1	1	0	0	0
2	2	2	4	5
3	12	9	9	8
Unknown	3	1	1	4
ER status, n				
Negative	18	12	14	17
Positive	0	0	0	0
PgR status, n				
Negative	17	12	13	17
Positive	1	0	1	0
HER2 status (IHC), n				
0	17	10	11	12
1+	1	2	2	3
2+	0	0	1	2
3+	0	0	0	0
Ki67 index, %				
Median (range)	58.6 (20.2, 92.4)	64.2 (36.4, 86)	54.7 (20, 90.4)	56.2 (16.2, 82)
<30, n	3	0	4	4
≥30, n	15	12	10	13
SpCRypN0, n				
Yes	12	3	2	4
No	6	9	12	13
HRD score, n				
<42	0	0	5	9

≥42	18	12	0	0
NA	0	0	9	8
Treatment arm, n				
A1	18	-	-	-
A2	-	12	-	-
B1	-	-	14	-
B2	-	-	-	17

Abbreviations: ER, estrogen receptor; PgR, progesterone receptor; HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; SpCRypN0, strict pathological complete response, metastasis in axillary lymph nodes ≤ 0.2 mm; HRD, homologous recombination deficiency; NA, not available.

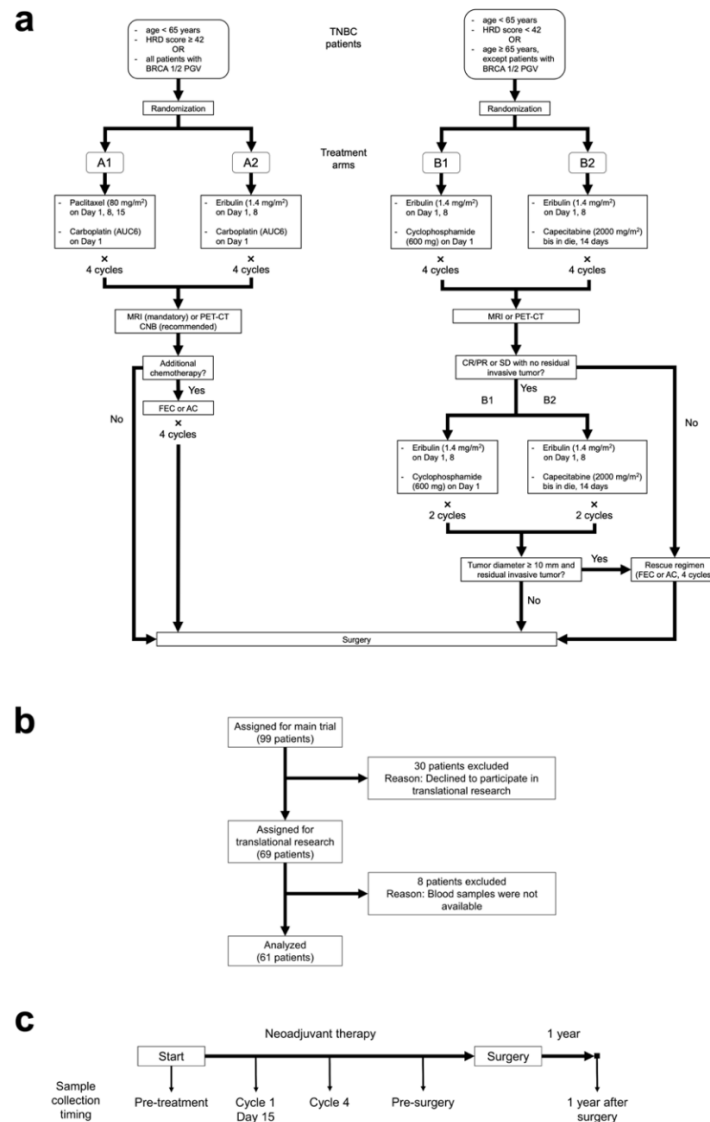


Fig. 1 Summary of this clinical trial. (a) Flowchart showing the process of the clinical trial from which the data were obtained (13). HRD, homologous recombination deficiency; PGV, pathogenic germline variant; MRI, magnetic resonance imaging; PET-CT, positron emission tomography-computed tomography; CNB, core needle biopsy; FEC, 5-fluorouracil–epirubicin–cyclophosphamide; AC, doxorubicin-cyclophosphamide; CR, complete response; PR, partial response; SD, stable disease. **(b)** Flowchart showing the process of screening the participants of the study. **(c)** Flowchart showing the process of blood sample collection from the participants.

2.5 Investigating the biological significance of cytokines using STRING DB

The previously discussed analysis methods (WGCNA, statistical analyses of individual cytokines) yielded a group of cytokines with potentially important correlations with clinical data. To gain a better understanding of the biological roles played by such cytokines, we utilized STRING DB, a publicly accessible database of proteins and their functional interactions [21]. STRING DB can infer protein-protein interaction (PPI) networks using a variety of evidence sources, such as experimental evidence, curated databases, and text mining. STRING DB considers not only direct protein interactions (i.e., proteins that physically bind with each other) but also indirect protein interactions (e.g., proteins that transcriptionally regulate each other, proteins that share the same substrate in a metabolic pathway, proteins that participate in larger multi-protein assemblies) [21]. The user inputs a single protein or a set of multiple proteins and specifies a set of settings, such as the sources of evidence to be used (e.g., experimental evidence, curated databases, text mining, etc.), the threshold confidence score for a predicted PPI to be displayed in the output graph, the maximum number of interactors to be displayed, etc. The outputs include a graph depicting the predicted PPIs, lists of Gene Ontology (GO) terms enriched in the enriched PPI network, a list of KEGG pathways enriched in the predicted PPI network, etc.

In this study, we used STRING DB version 11.5 [22][23]. We set the network type to “full STRING network”. We set the threshold interaction score to 0.700 and the maximum number of interactors to 20 for the first shell and 0 for the second shell. For the active interaction sources, we checked “Experiments” only. We set the meaning of the network edges to indicate confidence (i.e., the thickness of the edges indicates the strength of data support). For the functional enrichment analysis, we set the statistical background to “whole genome.”

2.6 Area under the curve analysis

We endeavored to demonstrate the difference between the time series cytokine expression values of the patients with good clinical outcomes ($\text{SpCRypN0} = 1$) and the patients with poor clinical outcomes ($\text{SpCRypN0} = 0$). To this end, we performed an area under the curve (AUC) analysis for a certain set of cytokines to quantitatively compare the difference in

observed concentrations of these cytokines across all time points between the group with $SpCRypN0 = 1$ and the group with $SpCRypN0 = 0$. Using the observed cytokine concentrations at each time point, an area chart was plotted for each cytokine in the set for each of the two groups ($SpCRypN0 = 0$ and $SpCRypN0 = 1$). The total AUC of each cytokine for each group was calculated using GraphPad Prism version 9.4.1. We set the baseline to $Y = 0$ and the minimum peak height to 10% of the distance from the minimum to maximum Y . The direction of peaks was set such that all peaks must go above the baseline. The percent increase in AUC was calculated using Equation (1):

$$\% \text{ increase of AUC} = \frac{Total\ AUC_{SpCRypN0=1} - Total\ AUC_{SpCRypN0=0}}{Total\ AUC_{SpCRypN0=0}} \times 100 \quad (1)$$

2.7 Prediction of treatment outcomes using logistic regression with L2 regularization

We investigated whether the cytokine expression values could be utilized as predictors of patient response to chemotherapy. For this purpose, we performed logistic regression with L2 regularization to test whether or not pre-treatment cytokine expression levels could predict treatment outcomes. Logistic regression with L2 regularization was performed using the scikit-learn package of Python [24]. We used pre-treatment cytokine expression values as the predictor variables for this analysis. The basic equation of logistic regression is given by Equation (2):

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k \quad (2)$$

where p indicates the probability of an object being in the positive class, k is the number of predictor variables, x are the predictor variables, and β are parameters whose values are fitted according to the training data. The coefficients $\beta_1, \dots, \beta_k$ represent the variable importance of $x_1, \dots, x_k$, respectively, with larger magnitudes of a coefficient indicating greater variable importance of the corresponding predictor variable.

We also wished to investigate whether the accuracy of predictions depended on the treatment regimen. Hence, logistic regression was performed for three cases: prediction of

the outcome of eribulin-based chemotherapy, prediction of the outcome of paclitaxel-based chemotherapy, and prediction of the outcome of chemotherapy in general. For the first case, the model was trained using the cytokine observed concentration values of the patients in the eribulin-based treatment arms (A2-arm, B1-arm, and B2-arm). For the second case, the model was trained using the cytokine observed concentration values of the patients in the paclitaxel-based treatment arm (A1-arm). For the third case, the model was trained using the cytokine observed concentration values of patients in all treatment arms.

To assess the performance of logistic regression with L2 regularization, we calculated the accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of the logistic regression model using the following equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (4)$$

$$Specificity = \frac{TN}{TN + FP} \quad (5)$$

$$PPV = \frac{TP}{TP + FP} \quad (6)$$

$$NPV = \frac{TN}{TN + FN} \quad (7)$$

...where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

We also wished to evaluate the generalizability of the prediction model. Having a generalizable model is important so that the model can make reliable predictions for samples outside of the training set. To evaluate the generalizability of the logistic regression models, we performed 3-fold stratified cross validation. In 3-fold stratified cross validation, the data set is divided into 3 parts, with each part containing objects of both the positive and negative classes (Fig. A2). In each fold, 2 parts are used as training data for the model, and the remaining 1 part is used as testing data (Fig. A2). The performance metrics (accuracy,

sensitivity, specificity, PPV, and NPV) were calculated for each fold, after which the average values of each metric over the 3 folds were calculated.

3. Correlations between cytokines and TNBC chemotherapy outcomes

3.1 Correlation between cytokines and clinical phenotypes discovered by WGCNA

We generated datasets for the concentrations of 69 cytokines measured at five different time points from 61 patients with TNBC (Table 1: a total of 294 samples). WGCNA was performed using the averaged data from the 1st and 2nd measurements that were taken using the 48-plex and 40-plex kits. WGCNA identified ten distinct modules, defined as clusters of cytokines whose expression levels are correlated with each other (Fig. 2a). Correlations of particular interest are indicated with a sharp mark (#), and correlations with a q -value less than 0.1 are indicated with an asterisk (*) in Fig. 2a. Module 5 showed a relatively strong positive correlation with treatment outcome (SpCRypN0), with $r = 0.201$. Modules 1 and 10 showed relatively strong negative correlations with HRD scores, with $r = -0.239$ and $r = -0.241$, respectively. Modules 1 and 8 showed relatively strong positive correlations with patient ages, with $r = 0.349$ and $r = 0.205$, respectively. The cytokines belonging to each of the aforementioned modules are listed in Fig. 2b.

3.2 Identification of five cytokines whose expression positively correlates with treatment outcomes

Module 5 included 15 different cytokines. We next wanted to know which cytokines in this module contributed to the significant association between the module and treatment outcome. For this purpose, we investigated the correlations of 15 cytokines listed in module 5 with treatment outcomes individually. We compared the expression of individual cytokines in SpCRypN0 = 0 and SpCRypN0 = 1 and calculated q -values. As a result, among the cytokines in module 5 (Fig. 2b), the expression levels of IL-1 α , TRAIL, SCF, RANTES, and IL-16 were significantly correlated with treatment outcome ($q < 0.1$) (Fig. 3). These five cytokines had higher median observed concentration values in patients with favorable treatment outcomes (SpCRypN0 = 1) compared with patients with unfavorable treatment outcomes (SpCRypN0 = 0). Based on this, we concluded that the five cytokines contributed to the significant association between module 5 and treatment outcome. This result also suggested that further investigation of these five cytokines may provide insights into a mechanism underlying differences in treatment response.

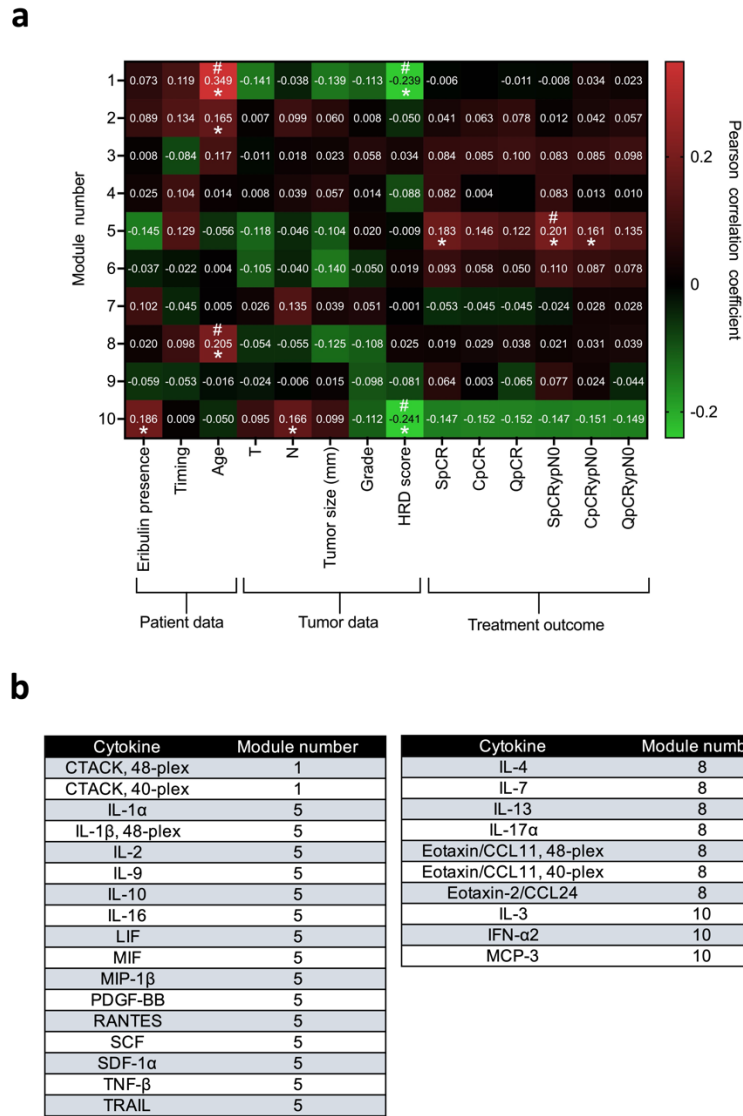


Fig. 2 Association among observed cytokine concentrations, clinical outcomes, HRD scores, and patient ages. (a) Heatmap showing the Pearson correlation coefficients between the clinical characteristics and the module eigengenes produced by WGCNA. Asterisks (*) indicate the correlations considered as discoveries ($q < 0.1$), and hashtags (#) indicate the correlations that are emphasized. Timing refers to the timing of blood sample collection. T and N refer to the tumor and lymph node numbers, respectively, of the TNM cancer staging system. HRD, homologous recombination deficiency score. **(b)** Table showing the cytokines included in the modules corresponding to the correlations marked with “#” in panel a.

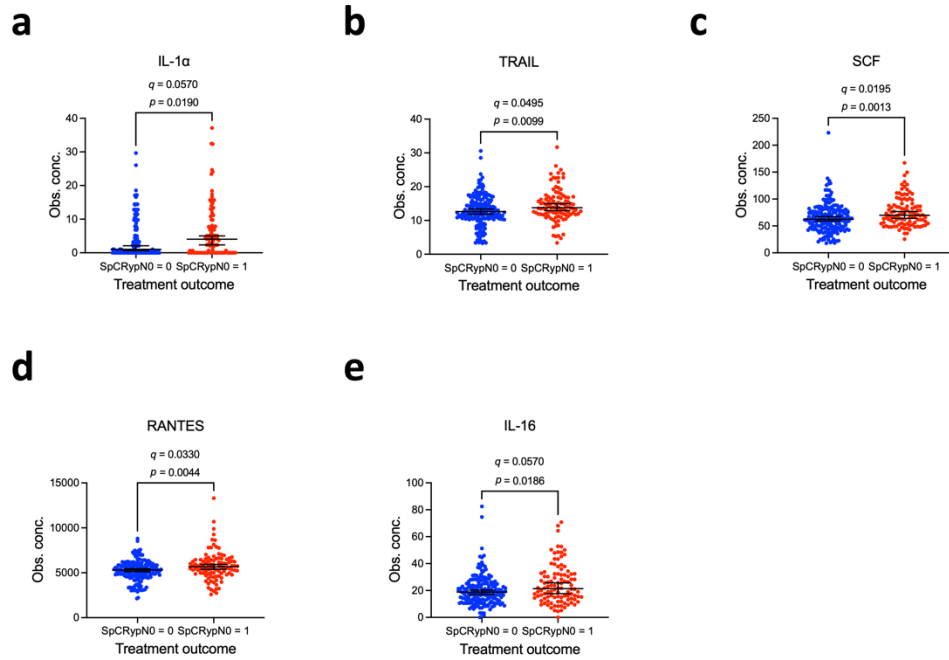


Fig. 3 Cytokines correlated with clinical outcomes. Categorical scatter plots showing the distribution of observed concentration (Obs. conc.) values of IL-1 α (a), TRAIL (b), SCF (c), RANTES (d), and IL-16 (e) across all time points for patients with SpCRypN0 = 0 and 1. Bars represent medians with 95% confidence intervals. p , p -value from the Mann-Whitney U -test; q , q -value from the method of Benjamini and Hochberg (1995).

3.3 The five cytokines show a similar expression pattern during treatment

In Fig. 3, we combined the data from all sampling points. Despite no significant association between the modules and “Timing” found in our WGCNA analysis (Fig. 2), we wanted to know how the five cytokines in module 5 were differentially expressed during treatment. To address this, we visualized the changes in the expression levels of these cytokines across multiple time points. Fig. 4 shows the mean observed concentrations of IL-1 α , TRAIL, SCF, RANTES, and IL-16 at each time point. We performed an AUC analysis of the mean observed concentrations of these five cytokines at each time point to test whether the constitutively high expression of these cytokines at multiple time points was correlated with treatment outcome. The percent increase in the AUC was calculated according to Equation (1) in the Methods section. The AUC analysis revealed that all five cytokines had a higher AUC in patients with favorable treatment outcomes (SpCRypN0 = 1) compared with patients with unfavorable treatment outcomes (SpCRypN0 = 0), i.e., % increase of AUC > 0 for all five cytokines (Fig. 4). Together with Fig. 3, these results indicate that constitutively high expression of IL-1 α , TRAIL, SCF, RANTES, and IL-16 at multiple time points during chemotherapy is correlated with favorable chemotherapy treatment outcomes in TNBC patients.

In the analysis, we noted that the expression patterns of these five cytokines appeared somewhat similar. The observed concentrations of IL-1 α , TRAIL, SCF, RANTES, and IL-16 tended to decrease during cycle 1-day 15 and cycle 4 compared with pre-treatment levels in both patient groups. In particular, in patients with favorable clinical outcomes, we found that IL-1 α , TRAIL, RANTES, and IL-16 tended to elevate after cycle 4. These results indicate that IL-1 α , TRAIL, SCF, RANTES, and IL-16 were down-regulated due to chemotherapy. It is of note that our search for cytokines associated with treatment outcomes led to the identification of cytokines that were likely affected by chemotherapy.

Based on the downregulation of IL-1 α , TRAIL, SCF, RANTES, and IL-16 during chemotherapy and the correlation between constitutively high expression of these five cytokines and favorable treatment outcomes, we hypothesize that IL-1 α , TRAIL, SCF, RANTES, and IL-16 are involved in the anti-tumor immune response during chemotherapy in TNBC patients. Elimination of cancer cells during chemotherapy requires cytokine-dependent reprogramming of the tumor immune microenvironment immune cell activity [20]. Given this, the downregulation of these five cytokines during chemotherapy might have been caused by chemotherapy-induced damage to the immune system, potentially reducing the efficacy of immune cell-dependent cancer cell elimination. Patients with constitutively high expressions of IL-1 α , TRAIL, SCF, RANTES, and IL-16 were thought to be more resistant to such chemotherapy-induced downregulation of the immune system, having a stronger anti-tumor immune response against breast cancers compared with patients with unfavorable treatment outcomes (SpCRypN0 = 0). At this point, the mechanisms underlying constitutively high expression of these five cytokines remain unknown.

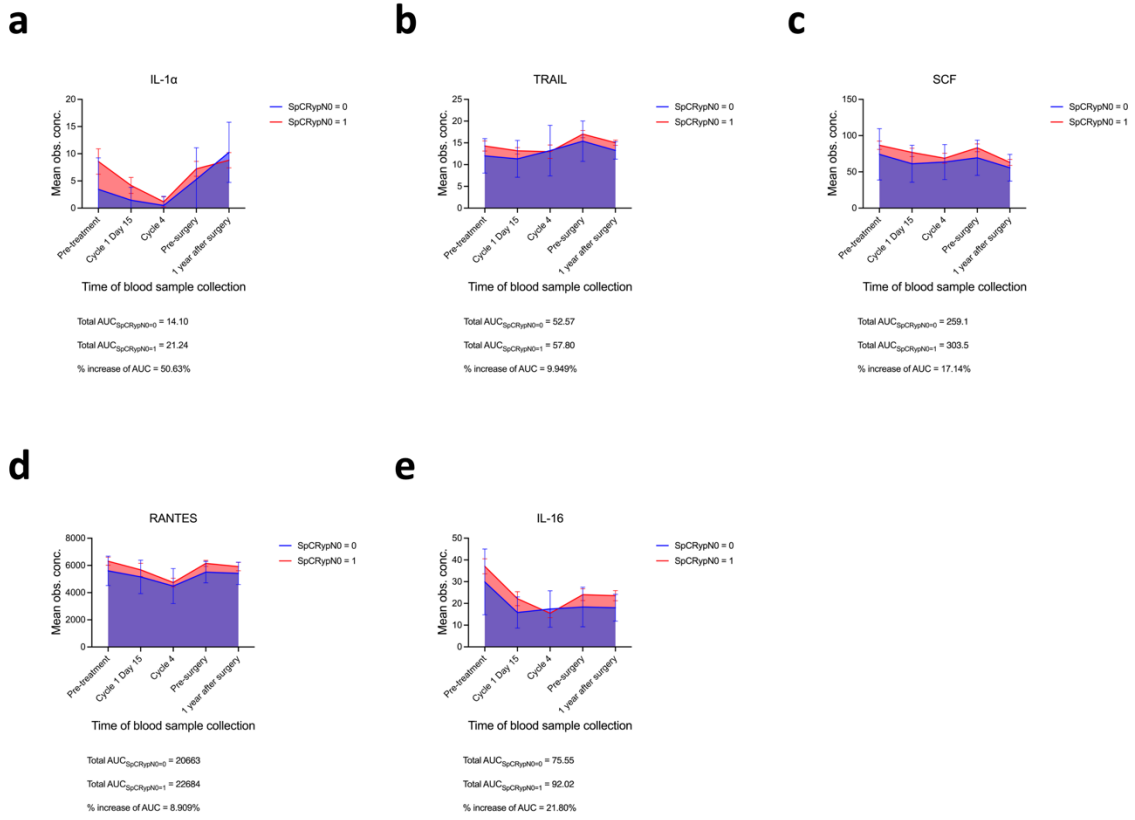


Fig. 4 Area under the curve (AUC) analysis of IL-1 α , TRAIL, SCF, RANTES, and IL-16. Area graphs of the mean observed concentrations of IL-1 α (a), TRAIL (b), SCF (c), RANTES (d), and IL-16 (e) across all time points. The red areas in the graphs represent patients with SpCRypN0 = 1, and the blue areas represent patients with SpCRypN0 = 0. Red bars represent the standard error of the mean (SEM) for patients with SpCRypN0 = 1. Blue bars represent the SEM for patients with SpCRypN0 = 0. For each graph, total AUC_{SpCRypN0=1} is the total AUC of the red area, total AUC_{SpCRypN0=0} is the total AUC of the blue area, and % increase of AUC is computed according to Equation (1) given in the Methods section.

3.4 PPI analyses implicate roles of the five cytokines in immune regulation

To further test our hypothesis that the expression levels of the five cytokines may be useful for inferring the status of the immune cell activity, we performed a PPI analysis using STRING DB, with the settings specified as described in the Methods section. The resulting PPI, which contains a total of 16 nodes (proteins), is shown in Fig. 5a. We also used STRING DB to obtain a list of Gene Ontology (GO) terms that are enriched in this PPI. The list of top 10 GO terms (biological process) based on the lowest false discovery rates is shown in Table 3. The numbers of proteins in the PPI that are labeled with each GO term are also given in Table 3.

As shown in Fig. 5b and Table 3, we found that 15 out of 16 proteins in this network are labeled with the term GO:0002376 (immune system process), and 8 out of 16 nodes are labeled with the term GO:0050900 (leukocyte migration). Overall, these results indicate that IL-1 α , TRAIL, SCF, RANTES, and IL-16 are involved in the functioning of the immune system. This provides further support to our hypothesis that constitutively high expressions of IL-1 α , TRAIL, SCF, RANTES, and IL-16 are a sign of strong anti-tumor immune response, which may contribute to the better clinical outcomes observed in patients with constitutively high expressions of these cytokines.

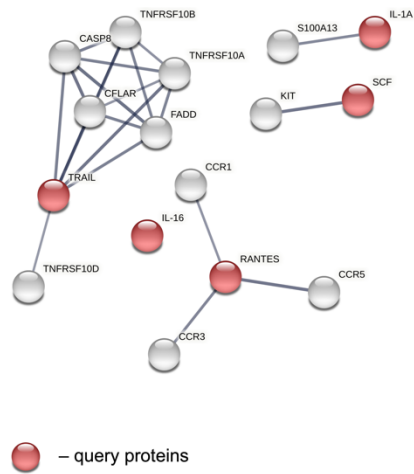
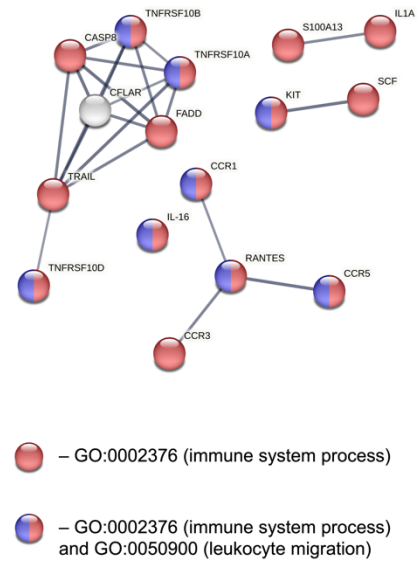
a**b**

Fig. 5 Protein-protein interaction (PPI) network analysis. (a) The PPI network of IL-1 α , TRAIL, SCF, RANTES, and IL-16 obtained using STRING DB version 11.5. Red nodes represent the five query proteins (IL-1 α , TRAIL, SCF, RANTES, and IL-16), and white nodes represent proteins that are predicted to interact with the five query proteins. Line thickness indicates the strength of data support of each predicted interaction. (b) The same PPI network colored according to GO terms (biological processes). Red color indicates proteins labeled with the GO term GO:0002376 (immune system process). Blue color indicates proteins labeled with the GO term GO:0050900 (leukocyte migration). Nodes with two colors indicate proteins labeled with both GO terms.

Table 3. Top 10 Gene Ontology (GO) terms (biological processes) in the PPI network in

Figure 5. The GO terms are arranged according to the false discovery rate.

GO term ID	GO term description	Observed count in network	False discovery rate
GO:0043069	Negative regulation of programmed cell death	12	2.12E-09
GO:0002376	Immune system process	15	3.25E-09
GO:1902042	Negative regulation of extrinsic apoptotic signaling pathway via death domain receptors	6	3.25E-09
GO:1902533	Positive regulation of intracellular signal transduction	12	3.25E-09
GO:2001237	Negative regulation of extrinsic apoptotic signaling pathway	7	4.14E-09
GO:0043066	Negative regulation of apoptotic process	11	1.24E-08
GO:0007166	Cell surface receptor signaling pathway	14	1.57E-08
GO:0036462	TRAIL-activated apoptotic signaling pathway	4	6.90E-08
GO:0050900	Leukocyte migration	8	6.90E-08
GO:0006919	Activation of cysteine-type endopeptidase activity involved in apoptotic process	6	9.01E-08

4. Prediction of TNBC chemotherapy outcomes using a logistic regression model with L2 regularization

4.1 Basic principle of logistic regression models and L2 regularization

Logistic regression is a mathematical model which models the probability of an event by assuming that the log-odds of the event is a linear combination of a set of independent variables. As stated earlier, the basic equation of logistic regression is given in Equation (2). Logistic regression can be utilized to solve classification problems. To illustrate this concept, Equation (2) may be rewritten as:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)}} \quad (8)$$

where p indicates the probability of the event, k is the number of predictor variables, x are the predictor variables, and β are parameters whose values are fitted according to the training data. The coefficients $\beta_1, \dots, \beta_k$ represent the variable importance of $x_1, \dots, x_k$, respectively, with larger magnitudes of a coefficient indicating greater variable importance of the corresponding predictor variable. Thus, the probability of an event occurring is given by Equation (8). In our case, the event that we wish to model is a good treatment outcome of TNBC chemotherapy (i.e., SpCRypN0 = 1). If the probability of a good treatment outcome for a certain patient is above a certain threshold, then we classify that patient as SpCRypN0 = 1. Conversely, if the probability of a good treatment outcome for a certain patient is below the threshold, then we classify that patient as SpCRypN0 = 0 (i.e., poor treatment outcome). In this research, we use a threshold of probability of 0.5 (50%). Through this procedure, we can predict the treatment outcomes of patients using the logistic regression model.

To perform such calculations using Equation (8), it is necessary to determine the values of the coefficients $\beta_1, \dots, \beta_k$. In order to determine the optimal values of these coefficients, the logistic regression model utilizes a cost function. The cost function represents the cost, or “penalty”, which is incurred by making a wrong prediction (i.e., predicting that a patient will have a good treatment outcome when, in fact, the actual treatment outcome was poor). Intuitively, we would like to impose a high cost for making an inaccurate prediction, and a low cost for making an accurate prediction. Hence, the cost function is constructed from the

ground truth of our training data set. If the model's predictions mostly agree with the ground truth, then the cost will be low. Conversely, if the model's predictions mostly disagree with the ground truth, the cost will be high.

Given a training data set whose ground truth is known (in this case, a set of patients whose actual treatment outcomes are known), the cost can be calculated for each training data point. Suppose that we have a certain patient with a known treatment outcome. Let y be the treatment outcome and suppose that $y = 0$ represents a poor treatment outcome, and $y = 1$ represents a good treatment outcome. Let p be the probability of a good treatment outcome as given by Equation (8). Let j be the total number of samples in the training data set. For logistic regression, the cost for each i th training data point is given by the following equation:

$$Cost(p_i, y_i) = (-y_i * \ln(p_i)) - ((1 - y_i) * \ln(1 - p_i)) \quad (9)$$

...where p_i is the predicted probability of having a good treatment outcome for the i th patient, and y_i is the actual treatment outcome of the i th patient. Let C be the total cost of all the training data points. To calculate C , we simply take the average all the costs of the individual points. Then C is given by the following equation:

$$C = \frac{1}{j} \sum_{i=1}^j Cost(p_i, y_i) \quad (10)$$

The parameters $\beta_1, \dots, \beta_k$ are determined by minimizing C in the Equation (10). In addition to the basic cost function of Equation (10), it is also possible to introduce a regularization term. The purpose of the regularization term is to reduce the risk of overfitting imposing a penalty that prevents the coefficients of the predictor variable from becoming excessively large. The regularization method known as L2 regularization uses a regularization term of the form:

$$\frac{\lambda}{2j} \sum_{p=1}^k \beta_p^2$$

where λ is a parameter that controls the strength of the penalty, j is the number of samples in the training data set, k is the number of predictors, and β_p are the coefficients of the predictors. Hence, if L2 regularization is utilized, then the cost function can be written as follows:

$$C = \frac{1}{j} \sum_{i=1}^j \text{Cost}(p_i, y_i) + \frac{\lambda}{2j} \sum_{p=1}^k \beta_p^2 \quad (11)$$

Thus, the optimal values of the coefficients $\beta_1, \dots, \beta_k$ are the values which minimize C as given by Equation (11).

4.2 Determination of cytokines which are strong predictors of treatment outcome

We next questioned whether our datasets obtained at pre-treatment could be a predictor for treatment outcomes, which is an important application of time-series data. To this end, we employed logistic regression to reveal whether or not expression levels of five cytokines at pre-treatment could be used as predictors of treatment outcomes. Logistic regression is a machine learning method that can be used for binary classification (i.e., classifying a set of objects into one of two categories). In this case, we used logistic regression to classify TNBC patients into SpCRypN0 = 0 (poor treatment outcome) or SpCRypN0 = 1 (good treatment outcome). Logistic regression analysis was first performed using the entire patient cohort, and using the five cytokines IL-1 α , TRAIL, SCF, RANTES, and IL-16 as predictor variables. Upon performing such analysis, the coefficients of the predictor variables ($\beta_1, \dots, \beta_k$ in Equation (2)) were determined to be $\beta_1 = 0.06465389$, $\beta_2 = 0.03733072$, $\beta_3 = -0.00559288$, $\beta_4 = 0.00017136$, and $\beta_5 = 0.01026023$. The coefficients $\beta_1, \beta_2, \beta_3, \beta_4$, and β_5 represent the variable importance of IL-1 α , TRAIL, SCF, RANTES, and IL-16, respectively. The relatively small magnitudes of β_3, β_4 , and β_5 indicate the relatively low importance of SCF, RANTES, and IL-16 in predicting the treatment outcomes. Hence, we simplified the logistic regression model by using IL-1 α and TRAIL as predictor variables.

Logistic regression analysis using IL-1 α and TRAIL as predictor variables for the patients in the eribulin-based treatment arms (i.e., treatment arms A2, B1, and B2) and the paclitaxel-based treatment arm (i.e., treatment arm A1) demonstrated that the expression levels of IL-1 α and TRAIL at pre-treatment can be a predictor of treatment outcomes in the eribulin arm

(Fig. 6). The accuracy in the eribulin-based treatment arms was 86.05%, which was higher than that in the paclitaxel arm (72.22%) (Fig. A3). In addition, the high values of positive predictive value (PPV) (100%) and negative predictive value (NPV) (85.00%) in the eribulin-based treatment arms (Fig. 6b) indicate that IL-1 α and TRAIL expression at pre-treatment can predict both positive (SpCRypN0 = 1) and negative (SpCRypN0 = 0) treatment outcomes with a high degree of confidence. Furthermore, the logistic regression model of eribulin-based chemotherapy showed high accuracy (83.65%) under 3-fold cross validation (Fig. 6 and Fig. A2), suggesting that the model is generalizable. The excellent performance of the logistic regression model in the eribulin-based treatment arm (Fig. 6) compared to the paclitaxel-based treatment arm (Fig. A3) and the combined cohort (Fig. A4) suggests that IL-1 α and TRAIL may play crucial roles in patient response to eribulin-based chemotherapy, but not in paclitaxel-based chemotherapy. Supporting this notion, we found that eribulin treatment down-regulated IL-1 α less than paclitaxel treatment did, and up-regulated TRAIL more than paclitaxel treatment did (Fig. A5).

a

Logistic regression confusion matrix
(eribulin-based treatment arms)

		Prediction	
		SpCRypN0 = 0	SpCRypN0 = 1
Actual	SpCRypN0 = 0	True negative 34	False positive 0
	SpCRypN0 = 1	False negative 6	True positive 3

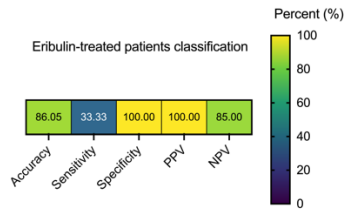
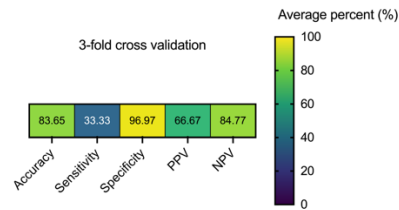
b**c**

Fig. 6 Logistic regression analysis of the patients in the eribulin-based treatment arms.

(a) Confusion matrix showing the results of prediction of treatment outcomes using a logistic regression model trained with the pre-treatment cytokine data of patients treated with eribulin-based chemotherapy. **(b)** Heatmap showing the values of accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of the prediction performed by the logistic regression model. **(c)** Heatmap showing the average values of accuracy, sensitivity, specificity, PPV, and NPV under 3-fold cross validation.

4.3 Performance of the prediction model

We next questioned whether our datasets obtained at pre-treatment could be a predictor for treatment outcomes, which is an important application of time-series data. To this end, we employed logistic regression to reveal whether or not expression levels of five cytokines at pre-treatment could be used as predictors of treatment outcomes. Logistic regression is a machine learning method that can be used for binary classification (i.e., classifying a set of objects into one of two categories). In this case, we used logistic regression to classify TNBC patients into $\text{SpCRypN0} = 0$ (poor treatment outcome) or $\text{SpCRypN0} = 1$ (good treatment outcome). Logistic regression analysis was first performed using the entire patient cohort, and using the five cytokines IL-1 α , TRAIL, SCF, RANTES, and IL-16 as predictor variables. Upon performing such analysis, the coefficients of the predictor variables ($\beta_1, \dots, \beta_k$ in Equation (2)) were determined to be $\beta_1 = 0.06465389$, $\beta_2 = 0.03733072$, $\beta_3 = -0.00559288$, $\beta_4 = 0.00017136$, and $\beta_5 = 0.01026023$. The coefficients β_1 , β_2 , β_3 , β_4 , and β_5 represent the variable importance of IL-1 α , TRAIL, SCF, RANTES, and IL-16, respectively. The relatively small magnitudes of β_3 , β_4 , and β_5 indicate the relatively low importance of SCF, RANTES, and IL-16 in predicting the treatment outcomes. Hence, we simplified the logistic regression model by using IL-1 α and TRAIL as predictor variables.

Logistic regression analysis using IL-1 α and TRAIL as predictor variables for the patients in the eribulin-based treatment arms (i.e., treatment arms A2, B1, and B2) and the paclitaxel-based treatment arm (i.e., treatment arm A1) demonstrated that the expression levels of IL-1 α and TRAIL at pre-treatment can be a predictor of treatment outcomes in the eribulin arm (Fig. 6). The accuracy in the eribulin-based treatment arms was 86.05%, which was higher than that in the paclitaxel arm (72.22%) (Fig. A3). In addition, the high values of positive predictive value (PPV) (100%) and negative predictive value (NPV) (85.00%) in the eribulin-based treatment arms (Fig. 6b) indicate that IL-1 α and TRAIL expression at pre-treatment can predict both positive ($\text{SpCRypN0} = 1$) and negative ($\text{SpCRypN0} = 0$) treatment outcomes with a high degree of confidence. Furthermore, the logistic regression model of eribulin-based chemotherapy showed high accuracy (83.65%) under 3-fold cross validation (Fig. 6 and Fig. A2), suggesting that the model is generalizable. The excellent performance of the logistic regression model in the eribulin-based treatment arm (Fig. 6) compared to the

paclitaxel-based treatment arm (Fig. A3) and the combined cohort (Fig. A4) suggests that IL-1 α and TRAIL may play crucial roles in patient response to eribulin-based chemotherapy, but not in paclitaxel-based chemotherapy. Supporting this notion, we found that eribulin treatment down-regulated IL-1 α less than paclitaxel treatment did, and up-regulated TRAIL more than paclitaxel treatment did (Fig. A5).

5. Correlation between cytokines and other clinical features

5.1 Cytokines correlated with HRD score

Aside from the the five cytokines associated with clinical outcomes, which was the focus of the previous chapter, we found other potentially important associations. Modules 1 and 10 had relatively strong negative correlations with HRD scores (Figure X). Previous research has indicated that HRD score is associated with the response to platinum and PARP inhibitors, but the association between HRD score and cytokine profiles is not yet well-understood [25][26]. Upon performing Pearson correlation analysis between the observed concentrations of the cytokines belonging to these two modules and the HRD scores of the patients in the cohort, the cytokines CTACK and IFN- α 2 were considered as discoveries ($q < 0.1$) (Fig. A6). CTACK was included in both the 48-plex and 40-plex kits (Table B2), and the averaged measurements from both kits passed the filtering procedure described in the Methods section (Table B3). Fig. A6a and Fig. A6b show the Pearson correlation analysis results for CTACK observed concentrations measured using the 48-plex and 40-plex kits, respectively. The observed concentration values of CTACK (48-plex), CTACK (40-plex), and IFN- α 2 were found to be negatively correlated with HRD scores, with Pearson correlation coefficients $r = -0.2831$, $r = -0.1608$, and $r = -0.2234$, respectively (Fig. A6a, b, c).

5.2 Cytokines correlated with age

Moreover, modules 1 and 8 had relatively strong positive correlations with patient ages. Previous studies have shown that age is a significant factor in response to chemotherapy [27]. Upon performing Pearson correlation analysis between the observed concentrations of the cytokines belonging to these two modules and the ages of the patients in the cohort, the cytokines IL-4, Eotaxin/CCL11, and CTACK were considered as discoveries ($q < 0.1$). Eotaxin/CCL11 and CTACK were included in both the 48-plex and 40-plex kits (Table B2), and the averaged measurements of both kits passed the filtering procedure described in the Methods section (Table B2). As shown in Fig. A7, the observed concentration values of IL-4, CTACK, and Eotaxin/CCL11 were found to be positively correlated with patient ages, with $r = 0.2102$, $r = 0.3425$, $r = 0.3087$, $r = 0.3232$, and $r = 0.2585$, respectively.

In line with these results, previous studies have indicated that Eotaxin is associated with aging [28].

5.3 Correlation between HRD score and patient age

Moreover, patient age also showed a weakly linear but statistically significant correlation with the HRD score (Fig. A8). This result is also in line with a previous study showing an association between HRD score and age [26]. We also found a series of cytokines that correlated with treatment toxicity (Fig. A9). Collectively, these findings indicate that our data are in agreement with previous studies, lending credence to our current measurements. Thus, these results indicate that blood cytokine profiling is a powerful tool to understand the nature of cancers and patients' backgrounds.

6. Discussion

6.1 The immunological roles played by IL-1 α , TRAIL, SCF, RANTES, and IL-16

Our data demonstrated that constitutively high expression levels of IL-1 α , TRAIL, SCF, RANTES, and IL-16 are correlated with favorable clinical outcomes of patients on chemotherapy (SpCRypN0 = 1). The results of STRING DB analysis (Fig. 5) indicate that these five cytokines interact with various proteins related to immune system processes and leukocyte migration. Taking these results together, we hypothesize that these five cytokines together play a crucial role in regulating the anti-tumor immune response in TNBC patients. Indeed, a study by Araujo et al. indicated that expression of CCL5 (RANTES) was correlated with increased amounts of tumor-infiltrating lymphocytes (TILs) in TNBC patients, which is consistent with our PPI analysis results, which indicate that RANTES is labeled by the GO term GO:0050900 (leukocyte migration) [29]. Previous studies have also indicated that IL-1 α and TRAIL play roles in anti-tumor immune responses, although the exact role these cytokines play in TNBC is currently not well-known [30][31]. Although the roles played by SCF and IL-16 in the anti-tumor immune response are not yet well-understood, previous research has indicated that SCF can enhance innate immunity through its effects on mast cells and that IL-16 is associated with macrophage recruitment to breast cancer tumors [32][33]. These prior research results provide support to our hypothesis that constitutively high expressions of IL-1 α , TRAIL, SCF, RANTES, and IL-16 are a sign of a strong anti-tumor immune response. However, the mechanisms underlying such constitutive high expression of these cytokines remain currently unclear.

We also discovered that IL-1 α , TRAIL, SCF, RANTES, and IL-16 followed a pattern where their mean observed concentrations are decreased during chemotherapy and increased again after chemotherapy (Fig. 4).

6.2 Implications of the results of the prediction model

We also demonstrated, using logistic regression analyses, that IL-1 α and TRAIL expression at pre-treatment can be a predictor of treatment outcomes in the eribulin arm. This result suggests that logistic regression analysis of time-series cytokine expression data may be an effective strategy for precision chemotherapy for TNBC patients, helping clinicians decide

whether or not it is appropriate to use eribulin-based chemotherapy. Treatment options for TNBC are rapidly evolving, with therapies such as antibody-drug conjugates and immunotherapies increasingly being integrated into standard treatment regimens [34][35]. The increasing number of treatment options has increased the complexity of TNBC treatments, demanding novel methods to inform treatment decisions and improve patient outcomes. Our time series cytokine analyses match this clinically important demand, as it sheds light on the positioning of eribulin therapy based on the individual characteristics of each patient. The results of the logistic regression analysis highlight the utility of machine learning models for increasing the precision of cancer chemotherapy. Rigorous validation of the performance of such models is an important venue of future research, which may pave the way towards precision TNBC chemotherapy.

6.3 Synergistic effects of multiple cytokines

Another point that should be considered is the synergistic effects of multiple cytokines acting together. The majority of studies that have investigated the relationships between cytokines and cancer have focused only on the effects of one cytokine per experiment. This type of single-cytokine experiment is common due to the simplicity of performing such experiments and the ease of data interpretation. However, in the case of actual cancer patients, the tumor microenvironment contains many different cytokines. It is, therefore, likely that cytokines in the tumor microenvironment of cancer patients act synergistically in ways that cannot be replicated by single-cytokine experiments. Indeed, previous research investigating the synergistic effects of combinations of cytokines in the context of anti-viral immunity has shown that the effects of multiple cytokines acting together are not simply the summation of the effects of each individual cytokine [36][37]. For example, Barteet et al. demonstrated that treatment of human fibroblasts with TNF and IFN- β resulted in the upregulation of more than 800 genes that are not upregulated by either TNF or IFN- β alone [38]. Therefore, it is possible that constitutively high expressions of IL-1 α , TRAIL, SCF, RANTES, and IL-16 at the same time cause synergistic effects that are not simply the summation of the individual effects of each of these cytokines. The possibility of such synergistic effects is further highlighted by the results of the logistic regression analysis, which demonstrated that IL-1 α and TRAIL

could predict the outcome of eribulin-based chemotherapy in TNBC patients. This suggests that IL-1 α and TRAIL may act synergistically in the context of eribulin-based TNBC chemotherapy in a way that influences the treatment outcome. Addressing such synergistic effects is challenging but an important next venue of this research area.

6.4 Limitations of our approach

As mentioned in Section 6.1, we also discovered that IL-1 α , TRAIL, SCF, RANTES, and IL-16 followed a pattern where their mean observed concentrations are decreased during chemotherapy and increased again after chemotherapy (Fig. 4). This is despite the fact that no significant correlations ($q < 0.1$) were detected between the WGCNA modules and the timing of blood sample collection (Fig. 2a). This phenomenon can be explained by the fact that the expression patterns of IL-1 α , TRAIL, SCF, RANTES, and IL-16 with respect to the timing of blood sample collection (Fig. 4) are nonlinear and non-unidirectional (i.e., the trend contains both increasing and decreasing parts). Our data analysis pipeline makes extensive use of Pearson correlation coefficients (in WGCNA and in the correlations between individual cytokines and clinical phenotypes), which is simple to implement and well-suited for discovering linear correlations or unidirectional correlations between cytokine expression and clinical phenotypes. However, analyses based on Pearson correlation coefficients may fail to detect nonlinear correlations and non-unidirectional patterns, such as the expression patterns shown in Fig. 4. This result highlights an important limitation of this data analysis pipeline.

7. Conclusion

Based on the results of this research, we can make a number of conclusions. First, the results indicate that the cytokines IL-1 α and TRAIL may be reliable predictors of patient response to eribulin-based TNBC chemotherapy. Thus, IL-1 α and TRAIL may be considered as potential biomarkers that may be utilized for precision TNBC chemotherapy. However, it is necessary to bear in mind that this research was conducted with a relatively small sample size (61 patients). Hence, further studies are needed in order to confirm the reliability of IL-1 α and TRAIL as biomarkers of precision TNBC chemotherapy.

Second, the results indicate that logistic regression with L2 regularization may be an effective method of predicting the results of eribulin-based TNBC chemotherapy using blood cytokine measurements. The ability of logistic regression with L2 regularization to identify the two cytokines IL-1 α and TRAIL as being particularly strong predictors of treatment outcome, and the high accuracy obtained when predicting the outcome of eribulin-based TNBC chemotherapy, highlight the utility of this method as a tool for discovering biomarkers for precision cancer therapy. However, as mentioned previously, the small sample size presents a significant limitation of our study. Hence, additional studies are needed in order to rigorously validate the effectiveness of our approach.

Third, the results suggest that the synergistic effects of multiple cytokines may be an important factor to consider. At present, there is a large amount of previous experimental research investigating the roles played by cytokines in cancer. However, the majority of such experiments focused on only one cytokine at a time. Currently, there is little research on the effects of multiple cytokines acting synergistically. Although single-cytokine experiments are relatively straightforward to perform and interpret, these do not provide us with the full, comprehensive picture of the roles that cytokines play in cancer treatment. The synergistic interactions between various cytokines are a key part of this picture, and only by understanding such complex interaction can we fully understand the entire picture of how cytokines affect cancer treatment. As shown in our results, machine learning methods such as WGCNA and logistic regression with L2 regularization may be useful in determining

which cytokines are correlated with good treatment outcomes, which can help us to identify putative cytokine-to-cytokine synergistic interactions. Such putative interactions can then be confirmed experimentally in future studies.

Overall, these results highlight the utility of blood cytokine measurements and machine learning methods for precision cancer therapy. As the demand for new methods of cancer treatment increases, it is likely that blood cytokines of cancer patients will become an increasingly important topic of investigation. It is also likely that the use of machine learning methods will become increasingly important in future cancer research. We believe that blood cytokines and machine learning may play a key role in the development of new therapies not only for TNBC, but possibly also other types of cancer, thus improving survival rates of cancer patients.

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This dissertation was the result of years of diligent, hard work not only by myself, but also by those who guided my research. I would like to express my gratitude to my supervisors, Professor Kazushi Ikeda and Associate Professor Shinpei Kawaoka, whose advice and expertise were crucial for my research. I would also like to thank Dr. Kosuke Kawaguchi, who was the principal investigator of the project upon which this thesis was based, and who organized the experiments which measured the cytokine expression values. I would also like to thank all those who participated in the JBCRG-22TR project for all their valuable contributions. I would also like to thank all the members of my thesis committee for all their helpful advice and support.

Furthermore, I could not have made it this far without those who supported me in my personal life. I would like to thank my parents for providing me emotional support during times when I was feeling overwhelmed with stress. I would also like to thank my friends in the Nara Institute of Science and Technology and other universities in the Kansai region for all the fun times and precious memories that we made, which provided me a much-needed break from the stress of life as a graduate student. I would also like to thank the members of the Sama-Sama Community in the Cathedral of the Immaculate Conception (Osaka), who provided me with a place where I could feel the warmth of a Filipino community while living away from my native country.

And of course, I thank God, for having guided, protected, and blessed me for my entire life, and for bestowing upon me the talents and skills that I used to perform this research.

Appendices

A. Appendix figures

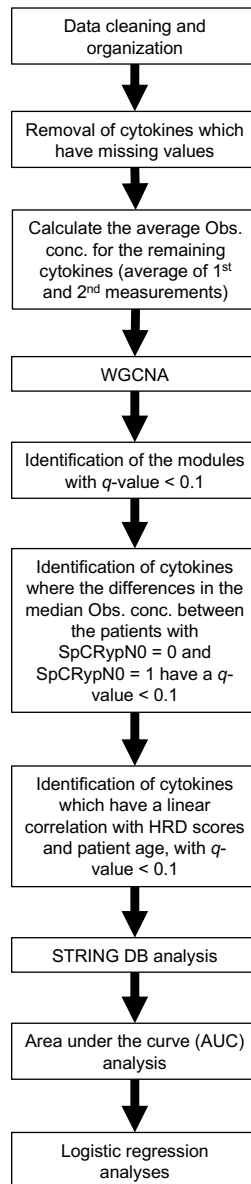


Fig. A1. Flowchart of the data analysis procedure.

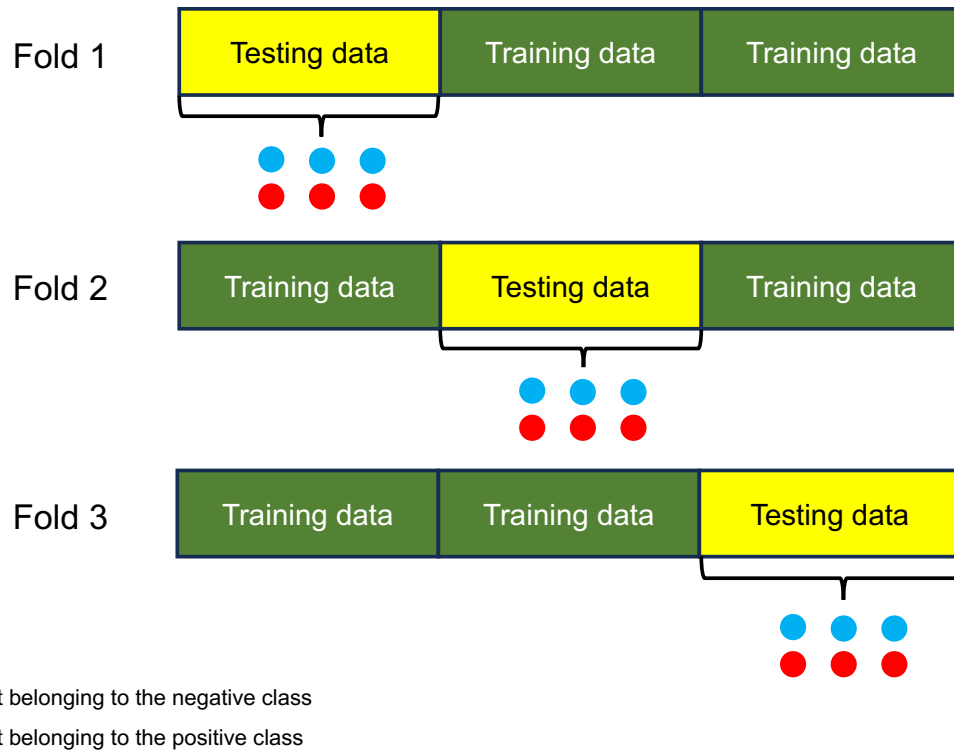


Fig. A2 An illustration of 3-fold cross validation. The data set is divided into 3 groups, each of which contains objects from both the positive and negative classes.

a

Logistic regression confusion matrix
(paclitaxel-based treatment arm)

		Prediction	
		SpCRypN0 = 0	SpCRypN0 = 1
Actual	SpCRypN0 = 0	True negative 1	False positive 5
	SpCRypN0 = 1	False negative 0	True positive 12

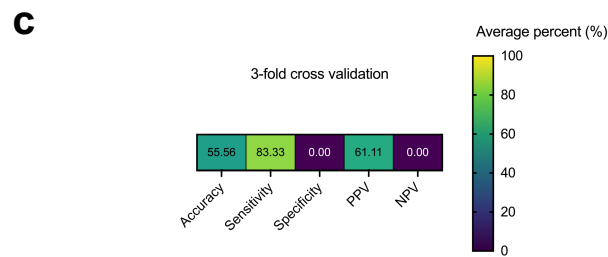
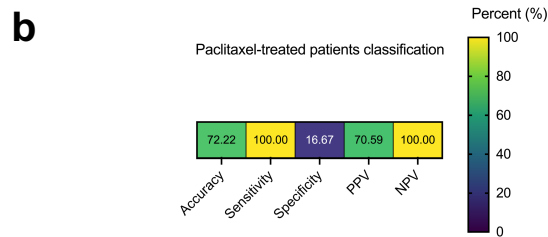


Fig. A3 Logistic regression analysis of the patients in the paclitaxel-based treatment arm. (a) Confusion matrix showing the results of prediction of treatment outcomes using a logistic regression model trained with the pre-treatment cytokine data of patients treated with paclitaxel-based chemotherapy. (b) Heatmap showing the values of accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of the prediction performed by the logistic regression model. (c) Heatmap showing the average values of accuracy, sensitivity, specificity, PPV, and NPV under 3-fold cross validation.

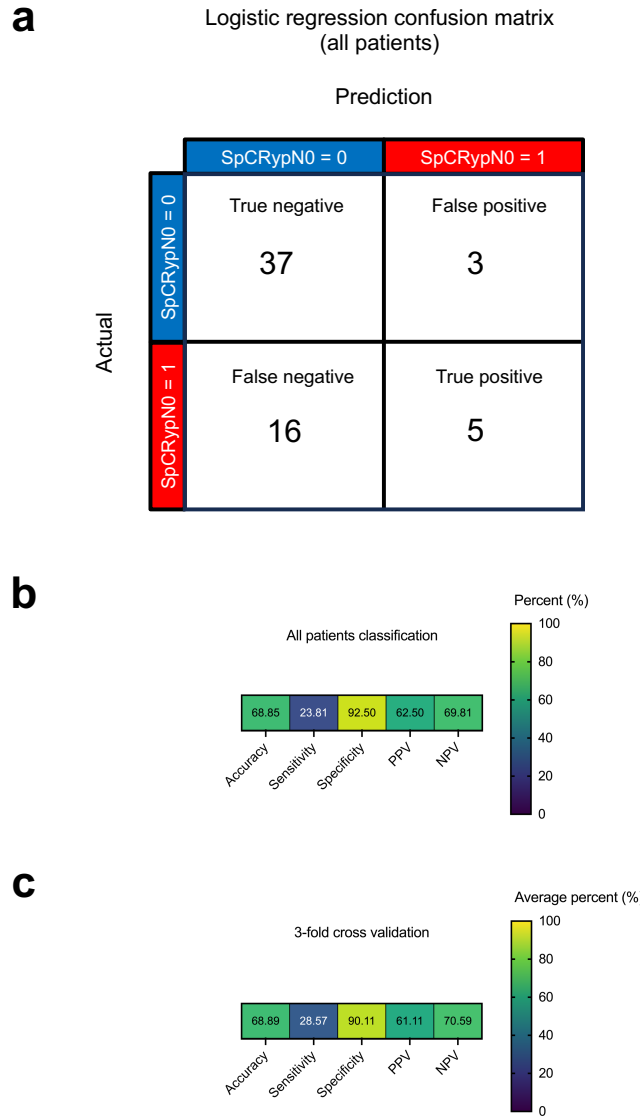


Fig. A4 Logistic regression analysis of the entire patient cohort. (a) Confusion matrix showing the results of prediction of treatment outcomes using a logistic regression model trained with the pre-treatment cytokine data of the entire patient cohort. (b) Heatmap showing the values of accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of the prediction performed by the logistic regression model. (c) Heatmap showing the average values of accuracy, sensitivity, specificity, PPV, and NPV under 3-fold cross validation.

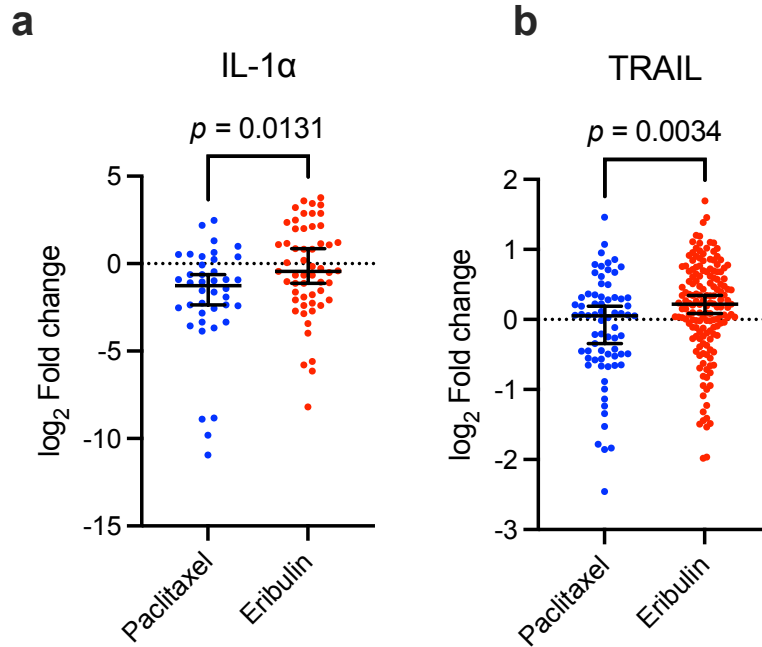


Fig. A5 Differential effects of paclitaxel-based and eribulin-based chemotherapy on the expression levels of IL-1 α and TRAIL. Categorical scatter plots showing the log₂ fold change of IL-1 α (a) and TRAIL (b). The log₂ fold change was calculated with respect to pre-treatment expression values. Points with zero expression values, and patients with zero pre-treatment expression values, were excluded due to the inability to calculate log₂ fold change for such cases. Bars represent medians with 95% confidence intervals. *p*, *p*-value from the Mann-Whitney *U*-test.

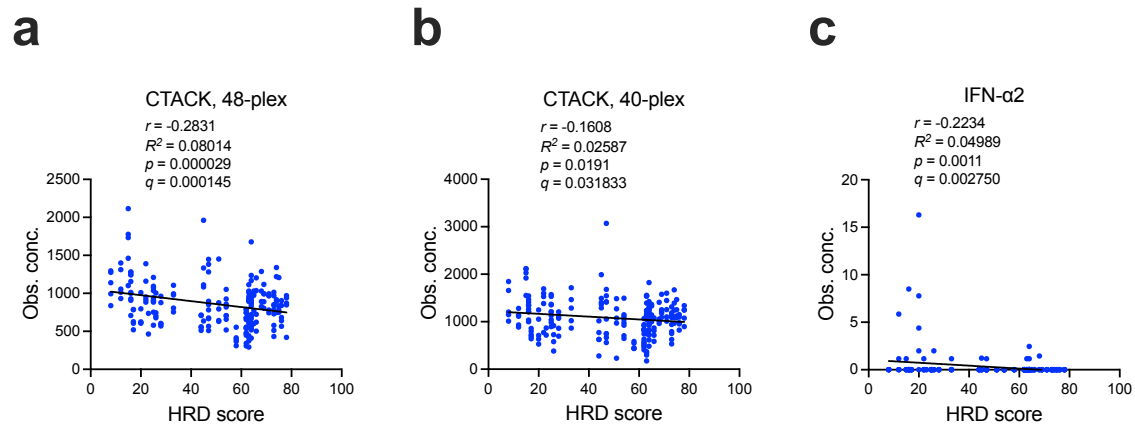


Figure A6. Correlations between cytokine observed concentrations and HRD scores.

Scatter plots showing the distribution of observed concentration (Obs. conc.) values of CTACK (48-plex kit) (**a**), CTACK (40-plex kit) (**b**), and IFN- $\alpha 2$ (**c**) in all patients across all time points versus HRD score, with the best-fit lines included. r , Pearson correlation coefficient; R^2 , coefficient of determination; p , p -value from Student's t -test; q , q -value from the method of Benjamini and Hochberg (1995).

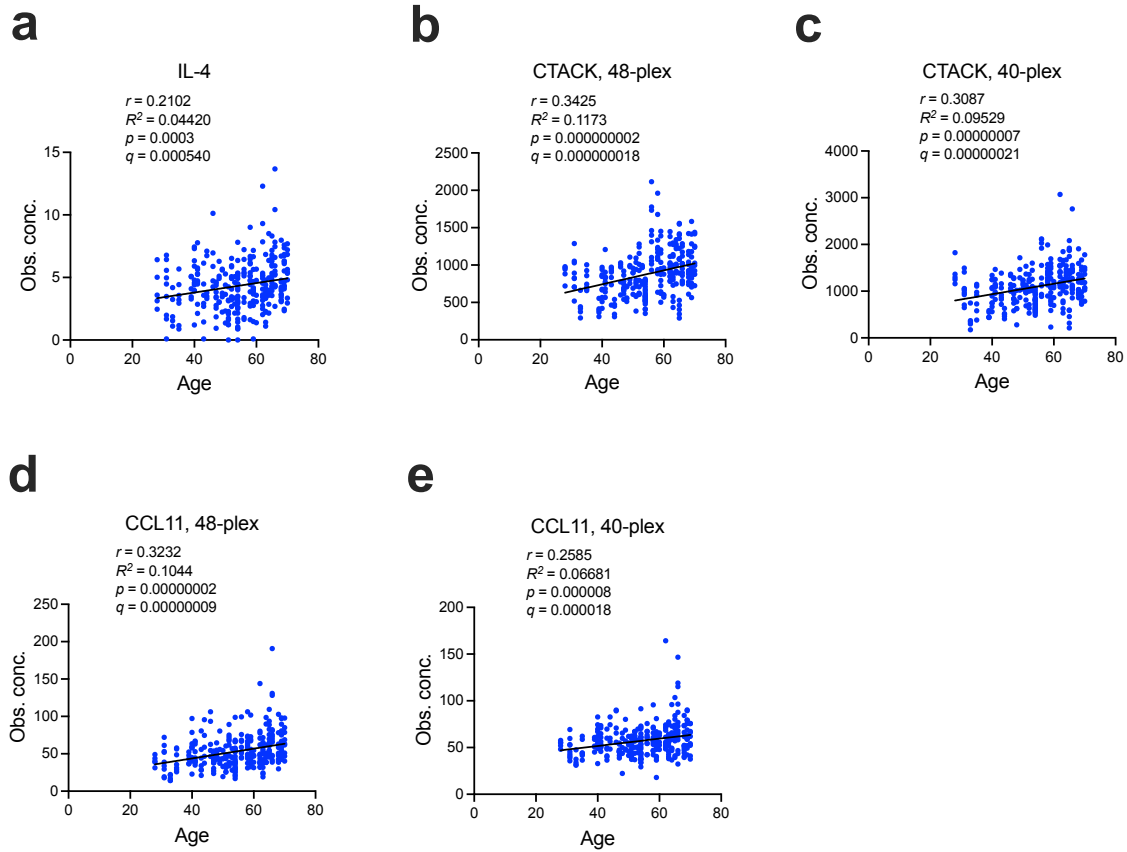


Figure A7. Correlations between cytokine observed concentrations and patient ages. Scatter plots showing the distribution of observed concentration (Obs. conc.) values of IL-4 (**a**), Eotaxin/CCL11 (48-plex kit) (**b**), Eotaxin/CCL22 (40-plex kit) (**c**), CTACK (48-plex kit) (**d**), and CTACK (40-plex kit) (**e**) in all patients across all time points versus patient ages, with best fit lines included. r , Pearson correlation coefficient; R^2 , coefficient of determination; p , p -value from Student's t -test; q , q -value from the method of Benjamini and Hochberg (1995).

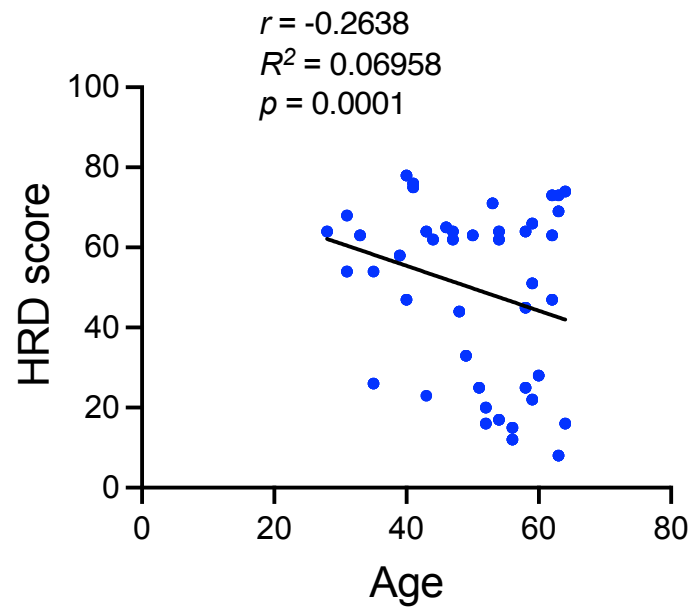


Figure A8. Correlation between patient ages and HRD scores. The line passing through the data points is the best fit line. r , Pearson correlation coefficient; R^2 , coefficient of determination; p , p -value from Student's t -test.

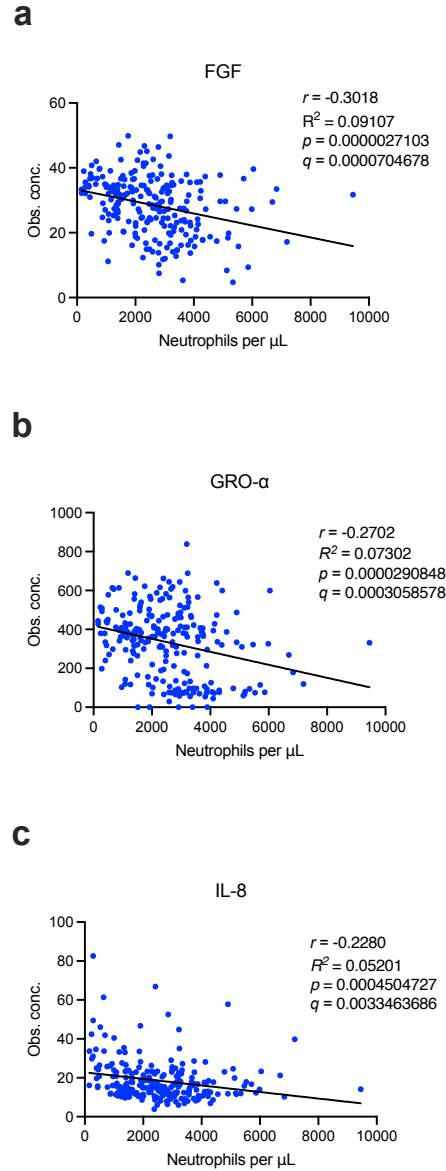


Fig. A9 Cytokines correlated with treatment toxicity. Scatter plots showing the observed concentration (Obs. conc.) values of FGF (**a**), GRO- α (**b**), and IL-8 (**c**) versus neutrophils per μL in each patient. Decrease in neutrophils was considered as a sign of treatment toxicity. Cytokines with q -values below 0.05 are shown in this figure. r , Pearson correlation coefficient; R^2 , coefficient of determination; p , p -value of the linear correlation; q , q -value from the method of Benjamini and Hochberg (1995).

B. Appendix tables

Table B1. Cytokines measured in this study.

Cytokine name	Abbreviation
β -nerve growth factor	β -NGF
B-lymphocyte chemoattractant-1	BCA-1
Chemokine (C-C motif) ligand 21	CCL-21
Cutaneous T cell-attracting chemokine	CTACK
Epithelial neutrophil activating peptide-78	ENA-78
Eotaxin	Eotaxin/CCL11
Eotaxin-2	-
Eotaxin-3	-
Fractalkine	-
Interleukin-1 α	IL-1 α
Interleukin-1 β	IL-1 β
Interleukin-1 receptor antagonist	IL-1 α
Interleukin-2	IL-2
Interleukin-2 receptor alpha chain	IL2R α
Interleukin-3	IL-3
Interleukin-4	IL-4
Interleukin-5	IL-5
Interleukin-6	IL-6
Interleukin-7	IL-7
Interleukin-8	IL-8

Interleukin-9	IL-9
Interleukin-10	IL-10
Interleukin-12 (p40)	IL-12 (p40)
Interleukin-12 (p70)	IL-12 (p70)
Interleukin-13	IL-13
Interleukin-15	IL-15
Interleukin-16	IL-16
Interleukin-17	IL-17
Interleukin-18	IL-18
Interferon gamma-induced protein 10	IP-10
Leukemia inhibitory factor	LIF
Granulocyte Macrophage Colony Stimulating Factor (GM-CSF)	GM-CSF
Granulocyte Colony-Stimulating Factor (G-CSF)	G-CSF
Macrophage Colony-Stimulating Factor (M-CSF)	M-CSF
Granulocyte Chemotactic Protein (GCP)-2	GCP-2
Interferon gamma	IFN- γ
Interferon alpha-2	IFN- α 2
Growth-regulated oncogene alpha	GRO- α
Growth-regulated oncogene beta	GRO- β

Hepatocyte growth factor	HGF
Chemokine (C-C motif) ligand 1	I-309/CCL1
IFN-inducible T-cell alpha chemoattractant	I-TAC
Monocyte chemoattractant protein-1	MCP-1
Monocyte chemoattractant protein-2	MCP-2
Monocyte chemoattractant protein-3	MCP-3
Monocyte chemoattractant protein-4	MCP-4
Macrophage-derived cytokine	MDC
Macrophage migration inhibitory factor	MIF
Monokine induced by gamma interferon	MIG
Macrophage inflammatory protein-1 α	MIP-1 α
Macrophage inflammatory protein-1 β	MIP-1 β
Macrophage inflammatory protein-1 δ	MIP-1 δ
Macrophage inflammatory protein-3 α	MIP-3 α
Macrophage inflammatory protein-3 β	MIP-3 β
Monocyte procoagulant inducing factor-1	MPIF-1
Platelet-derived growth factor-BB	PDGF-BB
Regulated Upon Activation Normal T-cell Expressed and Presumably Secreted	RANTES
Stem Cell Factor	SCF
Stem Cell Growth Factor Beta	SCGF- β
Chemokine (C-X-C motif) ligand 16	SCYB16

Stromal cell-derived factor 1 alpha	SDF-1 α
Stromal cell-derived factor 1 beta	SDF-1 β
Thymus-and-Activation-Regulated Chemokine	TARC
Thymus-Expressed Chemokine	TECK
Vascular endothelial growth factor A	VEGF-A
Basic fibroblast growth factor	bFGF
Tumor necrosis factor alpha	TNF- α
Tumor necrosis factor beta	TNF- β
TNF-related apoptosis inducing ligand	TRAIL

Table B2. List of cytokines included in the 48-plex and 40-plex kits.

48-plex cytokines	40-plex cytokines
Hu IL-1a (63)	Hu 6Ckine/CCL21 (12)
Hu IL-1b (39)	Hu BCA-1/CXCL13 (74)
Hu IL-1ra (25)	Hu CTACK/CCL27 (72)
Hu IL-2 (38)	Hu ENA-78/CXCL5 (73)
Hu IL-2Ra (13)	Hu Eotaxin/CCL11 (43)
Hu IL-3 (64)	Hu Eotaxin-2/CCL24 (30)
Hu IL-4 (52)	Hu Eotaxin-3/CCL26 (65)
Hu IL-5 (33)	Hu Fractalkine/CX3CL1 (77)
Hu IL-6 (19)	Hu GCP-2/CXCL6 (15)
Hu IL-7 (74)	Hu GM-CSF (34)
Hu IL-8 (54)	Hu Gro-a/CXCL1 (61)
Hu IL-9 (77)	Hu Gro-b/CXCL2 (78)
Hu IL-10 (56)	Hu I-309/CCL1 (20)
Hu IL-12(p70) (75)	Hu IFN-g (21)
Hu IL-12(p40) (28)	Hu IL-1b (39)
Hu IL-13 (51)	Hu IL-2 (38)
Hu IL-15 (73)	Hu IL-4 (52)
Hu IL-16 (27)	Hu IL-6 (19)
Hu IL-17A (76)	Hu IL-8/CXCL8 (54)
Hu IL-18 (42)	Hu IL-10 (56)

Hu CTACK (72)	Hu IL-16 (27)
Hu Eotaxin (43)	Hu IP-10/CXCL10 (48)
Hu FGF basic (44)	Hu I-TAC/CXCL11 (25)
Hu G-CSF (57)	Hu MCP-1/CCL2 (53)
Hu GM-CSF (34)	Hu MCP-2/CCL8 (57)
Hu GRO-a (61)	Hu MCP-3/CCL7 (26)
Hu HGF (62)	Hu MCP-4/CCL13 (28)
Hu IFN-a2 (20)	Hu MDC/CCL22 (29)
Hu IFN-g (21)	Hu MIF (35)
Hu IP-10 (48)	Hu MIG/CXCL9 (14)
Hu LIF (29)	Hu MIP-1a/CCL3 (55)
Hu MCP-1(MCAF) (53)	Hu MIP-1d/CCL15 (66)
Hu MCP-3 (26)	Hu MIP-3a/CCL20 (62)
Hu M-CSF (67)	Hu MIP-3b/CCL19 (76)
Hu MIF (35)	Hu MPIF-1/CCL23 (37)
Hu MIG (14)	Hu SCYB16/CXCL16 (64)
Hu MIP-1a (55)	Hu SDF-1a+b/CXCL12 (22)
Hu MIP-1b (18)	Hu TARC/CCL17 (67)
Hu b-NGF (46)	Hu TECK/CCL25 (46)
Hu PDGF-BB (47)	Hu TNF-a (36)
Hu RANTES (37)	
Hu SCF (65)	

Hu SCGF-b (78)	
Hu SDF-1a (22)	
Hu TNF-a (36)	
Hu TNF-b (30)	
Hu TRAIL (66)	
Hu VEGF-A (45)	

Table B3. Cytokines which passed the filtering procedure described in Section 2.3.

48-plex cytokines	40-plex cytokines
IL-1A	BCA-1/CXCL13
IL-1B	CTACK/CCL27
IL-2	Eotaxin/CCL11
IL-2RA	Eotaxin-2/CCL24
IL-3	Fractalkine/CX3CL1
IL-4	Gro- β /CXCL2
IL-5	IFN- γ
IL-6	IL-1 β
IL-7	IL-8/CXCL8
IL-9	
IL-10	
IL-12 (P70)	
IL-12 (P40)	
IL-13	
IL-15	
IL-16	
IL-17A	
IL-18	
CTACK	
EOTAXIN/CCL11	

FGF	
G-CSF	
GM-CSF	
GRO-A	
HGF	
IFN-A2	
IP-10	
LIF	
MCP-1 (MCAF)	
MCP-3	
M-CSF	
MIF	
MIG	
MIP-1A	
MIP-1B	
B-NGF	
PDGF-BB	
RANTES	
SCF	
SCGF-B	
SDF-1A	
TNF-B	

Bibliography

- [1] Kumar, P., & Aggarwal, R. (2016). An overview of triple-negative breast cancer. *Archives of gynecology and obstetrics*, 293(2), 247–269.
<https://doi.org/10.1007/s00404-015-3859-y>
- [2] Zagami, P., & Carey, L. A. (2022). Triple negative breast cancer: Pitfalls and progress. *NPJ breast cancer*, 8(1), 95. <https://doi.org/10.1038/s41523-022-00468-0>
- [3] Howard, F. M., & Olopade, O. I. (2021). Epidemiology of Triple-Negative Breast Cancer: A Review. *Cancer journal (Sudbury, Mass.)*, 27(1), 8–16.
<https://doi.org/10.1097/PPO.0000000000000500>
- [4] Zhu, S., Wu, Y., Song, B., Yi, M., Yan, Y., Mei, Q., & Wu, K. (2023). Recent advances in targeted strategies for triple-negative breast cancer. *Journal of hematology & oncology*, 16(1), 100. <https://doi.org/10.1186/s13045-023-01497-3>
- [5] Wu, S. Y., Fu, T., Jiang, Y. Z., & Shao, Z. M. (2020). Natural killer cells in cancer biology and therapy. *Molecular cancer*, 19(1), 120. <https://doi.org/10.1186/s12943-020-01238-x>
- [6] Durgeau, A., Virk, Y., Corgnac, S., & Mami-Chouaib, F. (2018). Recent Advances in Targeting CD8 T-Cell Immunity for More Effective Cancer Immunotherapy. *Frontiers in immunology*, 9, 14. <https://doi.org/10.3389/fimmu.2018.00014>
- [7] Liu, C., Chu, D., Kalantar-Zadeh, K., George, J., Young, H. A., & Liu, G. (2021). Cytokines: From Clinical Significance to Quantification. *Advanced science (Weinheim, Baden-Wurttemberg, Germany)*, 8(15), e2004433.
<https://doi.org/10.1002/advs.202004433>
- [8] Berraondo, P., Sanmamed, M. F., Ochoa, M. C., Etxeberria, I., Aznar, M. A., Pérez-Gracia, J. L., Rodríguez-Ruiz, M. E., Ponz-Sarvisé, M., Castañón, E., & Melero, I. (2019). Cytokines in clinical cancer immunotherapy. *British journal of cancer*, 120(1), 6–15. <https://doi.org/10.1038/s41416-018-0328-y>
- [9] Malone, M. K., Smrekar, K., Park, S., Blakely, B., Walter, A., Nasta, N., Park, J., Considine, M., Danilova, L. V., Pandey, N. B., Fertig, E. J., Popel, A. S., & Jin, K. (2020). Cytokines secreted by stromal cells in TNBC microenvironment as potential

- targets for cancer therapy. *Cancer biology & therapy*, 21(6), 560–569.
<https://doi.org/10.1080/15384047.2020.1739484>
- [10] Jabeen, S., Espinoza, J. A., Torland, L. A., Zucknick, M., Kumar, S., Haakensen, V. D., Lüders, T., Engebraaten, O., Børresen-Dale, A. L., Kyte, J. A., Gromov, P., Naume, B., Kristensen, V., Gromova, I., & Tekpli, X. (2018). Noninvasive profiling of serum cytokines in breast cancer patients and clinicopathological characteristics. *Oncoimmunology*, 8(2), e1537691.
<https://doi.org/10.1080/2162402X.2018.1537691>
- [11] Kawaguchi, K., Sakurai, M., Yamamoto, Y., Suzuki, E., Tsuda, M., Kataoka, T. R., Hirata, M., Nishie, M., Nojiri, T., Kumazoe, M., Saito, K., & Toi, M. (2019). Alteration of specific cytokine expression patterns in patients with breast cancer. *Scientific reports*, 9(1), 2924. <https://doi.org/10.1038/s41598-019-39476-9>
- [12] Groysman, L., Carlsen, L., Huntington, K. E., Shen, W. H., Zhou, L., & El-Deiry, W. S. (2021). Chemotherapy-induced cytokines and prognostic gene signatures vary across breast and colorectal cancer. *American journal of cancer research*, 11(12), 6086–6106.
- [13] Lehmann, B. D., Bauer, J. A., Chen, X., Sanders, M. E., Chakravarthy, A. B., Shyr, Y., & Pietenpol, J. A. (2011). Identification of human triple-negative breast cancer subtypes and preclinical models for selection of targeted therapies. *The Journal of clinical investigation*, 121(7), 2750–2767.
<https://doi.org/10.1172/JCI45014>
- [14] Lehmann, B. D., Jovanović, B., Chen, X., Estrada, M. V., Johnson, K. N., Shyr, Y., Moses, H. L., Sanders, M. E., & Pietenpol, J. A. (2016). Refinement of Triple-Negative Breast Cancer Molecular Subtypes: Implications for Neoadjuvant Chemotherapy Selection. *PloS one*, 11(6), e0157368.
<https://doi.org/10.1371/journal.pone.0157368>
- [15] Masuda, H., Baggerly, K. A., Wang, Y., Zhang, Y., Gonzalez-Angulo, A. M., Meric-Bernstam, F., Valero, V., Lehmann, B. D., Pietenpol, J. A., Hortobagyi, G. N., Symmans, W. F., & Ueno, N. T. (2013). Differential response to neoadjuvant chemotherapy among 7 triple-negative breast cancer molecular subtypes. *Clinical*

- cancer research : an official journal of the American Association for Cancer Research*, 19(19), 5533–5540. <https://doi.org/10.1158/1078-0432.CCR-13-0799>
- [16] Nurgali, K., Jagoe, R. T., & Abalo, R. (2018). Editorial: Adverse Effects of Cancer Chemotherapy: Anything New to Improve Tolerance and Reduce Sequelae?. *Frontiers in pharmacology*, 9, 245. <https://doi.org/10.3389/fphar.2018.00245>
- [17] Masuda, N., Bando, H., Yamanaka, T., Kadoya, T., Takahashi, M., Nagai, S. E., Ohtani, S., Aruga, T., Suzuki, E., Kikawa, Y., Yasojima, H., Kasai, H., Ishiguro, H., Kawabata, H., Morita, S., Haga, H., Kataoka, T. R., Uozumi, R., Ohno, S., & Toi, M. (2021). Eribulin-based neoadjuvant chemotherapy for triple-negative breast cancer patients stratified by homologous recombination deficiency status: a multicenter randomized phase II clinical trial. *Breast cancer research and treatment*, 188(1), 117–131. <https://doi.org/10.1007/s10549-021-06184-w>
- [18] Langfelder, P., & Horvath, S. (2008). WGCNA: an R package for weighted correlation network analysis. *BMC bioinformatics*, 9, 559. <https://doi.org/10.1186/1471-2105-9-559>
- [19] Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1), 289–300. <http://www.jstor.org/stable/2346101>
- [20] Dranoff G. (2004). Cytokines in cancer pathogenesis and cancer therapy. *Nature reviews. Cancer*, 4(1), 11–22. <https://doi.org/10.1038/nrc1252>
- [21] von Mering, C., Huynen, M., Jaeggi, D., Schmidt, S., Bork, P., & Snel, B. (2003). STRING: a database of predicted functional associations between proteins. *Nucleic acids research*, 31(1), 258–261. <https://doi.org/10.1093/nar/gkg034>
- [22] Szklarczyk, D., Gable, A. L., Lyon, D., Junge, A., Wyder, S., Huerta-Cepas, J., Simonovic, M., Doncheva, N. T., Morris, J. H., Bork, P., Jensen, L. J., & Mering, C. V. (2019). STRING v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic acids research*, 47(D1), D607–D613. <https://doi.org/10.1093/nar/gky1131>

- [23] Szklarczyk, D., Gable, A. L., Nastou, K. C., Lyon, D., Kirsch, R., Pyysalo, S., Doncheva, N. T., Legeay, M., Fang, T., Bork, P., Jensen, L. J., & von Mering, C. (2021). The STRING database in 2021: customizable protein-protein networks, and functional characterization of user-uploaded gene/measurement sets. *Nucleic acids research*, 49(D1), D605–D612. <https://doi.org/10.1093/nar/gkaa1074>
- [24] Pedregosa, F. *et al.* Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **12**, 2825-2830 (2011).
- [25] Chopra, N., Tovey, H., Pearson, A., Cutts, R., Toms, C., Proszek, P., Hubank, M., Dowsett, M., Dodson, A., Daley, F., Kriplani, D., Gevensleben, H., Davies, H. R., Degasper, A., Roylance, R., Chan, S., Tutt, A., Skene, A., Evans, A., Bliss, J. M., ... Turner, N. C. (2020). Homologous recombination DNA repair deficiency and PARP inhibition activity in primary triple negative breast cancer. *Nature communications*, 11(1), 2662. <https://doi.org/10.1038/s41467-020-16142-7>
- [26] Telli, M. L., Timms, K. M., Reid, J., Hennessy, B., Mills, G. B., Jensen, K. C., Szallasi, Z., Barry, W. T., Winer, E. P., Tung, N. M., Isakoff, S. J., Ryan, P. D., Greene-Colozzi, A., Gutin, A., Sangale, Z., Iliev, D., Neff, C., Abkevich, V., Jones, J. T., Lanchbury, J. S., ... Richardson, A. L. (2016). Homologous Recombination Deficiency (HRD) Score Predicts Response to Platinum-Containing Neoadjuvant Chemotherapy in Patients with Triple-Negative Breast Cancer. *Clinical cancer research : an official journal of the American Association for Cancer Research*, 22(15), 3764–3773. <https://doi.org/10.1158/1078-0432.CCR-15-2477>
- [27] Yu, J., Lin, C., Huang, J., Hong, J., Gao, W., Zhu, S., Lin, L., Chen, X., Huang, O., He, J., Zhu, L., Chen, W., Li, Y., Wu, J., & Shen, K. (2021). Efficacy of adjuvant chemotherapy stratified by age and the 21-gene recurrence score in estrogen receptor-positive breast cancer. *BMC cancer*, 21(1), 707. <https://doi.org/10.1186/s12885-021-08461-9>
- [28] Targowski, T., Jahnz-Rózyk, K., Plusa, T., & Glodzinska-Wyszogrodzka, E. (2005). Influence of age and gender on serum eotaxin concentration in healthy and allergic people. *Journal of investigational allergology & clinical immunology*, 15(4), 277–282.

- [29] Araujo, J. M., Gomez, A. C., Aguilar, A., Salgado, R., Balko, J. M., Bravo, L., Doimi, F., Bretel, D., Morante, Z., Flores, C., Gomez, H. L., & Pinto, J. A. (2018). Effect of CCL5 expression in the recruitment of immune cells in triple negative breast cancer. *Scientific reports*, 8(1), 4899. <https://doi.org/10.1038/s41598-018-23099-7>
- [30] Baker, K. J., Houston, A., & Brint, E. (2019). IL-1 Family Members in Cancer; Two Sides to Every Story. *Frontiers in immunology*, 10, 1197. <https://doi.org/10.3389/fimmu.2019.01197>
- [31] Cardoso Alves, L., Corazza, N., Micheau, O., & Krebs, P. (2021). The multifaceted role of TRAIL signaling in cancer and immunity. *The FEBS journal*, 288(19), 5530–5554. <https://doi.org/10.1111/febs.15637>
- [32] Möller, C., Alfredsson, J., Engström, M., Wootz, H., Xiang, Z., Lennartsson, J., Jönsson, J. I., & Nilsson, G. (2005). Stem cell factor promotes mast cell survival via inactivation of FOXO3a-mediated transcriptional induction and MEK-regulated phosphorylation of the proapoptotic protein Bim. *Blood*, 106(4), 1330–1336. <https://doi.org/10.1182/blood-2004-12-4792>
- [33] Richmond, J., Tuzova, M., Cruikshank, W., & Center, D. (2014). Regulation of cellular processes by interleukin-16 in homeostasis and cancer. *Journal of cellular physiology*, 229(2), 139–147. <https://doi.org/10.1002/jcp.24441>
- [34] Modi, S., Jacot, W., Yamashita, T., Sohn, J., Vidal, M., Tokunaga, E., Tsurutani, J., Ueno, N. T., Prat, A., Chae, Y. S., Lee, K. S., Niikura, N., Park, Y. H., Xu, B., Wang, X., Gil-Gil, M., Li, W., Pierga, J. Y., Im, S. A., Moore, H. C. F., ... DESTINY-Breast04 Trial Investigators (2022). Trastuzumab Deruxtecan in Previously Treated HER2-Low Advanced Breast Cancer. *The New England journal of medicine*, 387(1), 9–20. <https://doi.org/10.1056/NEJMoa2203690>
- [35] Schmid, P., Cortes, J., Dent, R., Pusztai, L., McArthur, H., Kümmel, S., Bergh, J., Denkert, C., Park, Y. H., Hui, R., Harbeck, N., Takahashi, M., Untch, M., Fasching, P. A., Cardoso, F., Andersen, J., Patt, D., Danso, M., Ferreira, M., Mouret-Reynier, M. A., ... KEYNOTE-522 Investigators (2022). Event-free Survival with Pembrolizumab in Early Triple-Negative Breast Cancer. *The New*

England journal of medicine, 386(6), 556–567.

<https://doi.org/10.1056/NEJMoa2112651>

- [36] Bartee, E., & McFadden, G. (2013). Cytokine synergy: an underappreciated contributor to innate anti-viral immunity. *Cytokine*, 63(3), 237–240.

<https://doi.org/10.1016/j.cyto.2013.04.036>

- [37] Mestan, J., Brockhaus, M., Kirchner, H., & Jacobsen, H. (1988). Antiviral activity of tumour necrosis factor. Synergism with interferons and induction of oligo-2',5'-adenylate synthetase. *The Journal of general virology*, 69 (Pt 12), 3113–3120. <https://doi.org/10.1099/0022-1317-69-12-3113>

- [38] Bartee, E., Mohamed, M. R., Lopez, M. C., Baker, H. V., & McFadden, G. (2009). The addition of tumor necrosis factor plus beta interferon induces a novel synergistic antiviral state against poxviruses in primary human fibroblasts. *Journal of virology*, 83(2), 498–511. <https://doi.org/10.1128/JVI.01376-08>

Publication list

[1] Saldajeno, D.P., Kawaoka, S., *et al.* (2024) Time-series blood cytokine profiles correlate with treatment responses in triple-negative breast cancer patients. *Br. J. Cancer*. DOI: <https://doi.org/10.1038/s41416-023-02527-0>, related to chapters 1 – 7 of the thesis